

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 513 — Past Questions & Materials  
**by Renance**

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Course: PEE 513  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 2  
Levels: 500

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## Paper 1 of 2 — PEE 513 (PETROLEUM PROPERTY EVALUATION)

Level 500 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN**  
**COLLEGE OF TECHNOLOGY**  
**DEPARTMENT OF PETROLEUM ENGINEERING**  
**2019/2020 First Semester Examination**  
**PNG 513: Petroleum Property Evaluation**

**Answer All Questions**

**Time: 2 Hours**

**Please write your Matriculation Number Boldly on your Script**

1) What is Petroleum Property Evaluation? Explain why it is essential in the Petroleum Industry?

- b) An oil field transfer pump having a useful life of five years is purchased for \$10,000. The pump will generate an incremental \$15,000 in the first year. However the revenue will decrease \$1000 per year in each of the next four years. Operating costs will be \$5000 in the first year but will increase \$1000 per year in each of the following four years. The asset owner is exempted from paying any taxes on this investment.
- Calculate the Financial net income for each year using straight line depreciation noting no salvage value
  - Determine the Undiscounted Profit to Investment Ratio using the Cash Flow Model.
  - Determine the Discounted Profit to Investment Ratio assuming that the asset owner has 10% earning power.

2) A management group has an exploratory budget of \$12.0 million for the fiscal year. They are considering drilling an exploratory well whose various possible outcomes are shown below. Operating profit can be taken as \$20/bbl and a PV factor of 0.7 may be assumed throughout the range of various possible reserve discoveries. If successful, the well will be completed with \$200,000 cost over the cost of a dry well. Management does not wish to exceed 5% risk. Based on risk alone, what is the maximum amount that could be paid as speculative cost, consisting of drilling costs and bonus paid for the lease.

| Probability | Possible Reserves (bbl) X 10 <sup>3</sup> |
|-------------|-------------------------------------------|
| 0.2         | 0                                         |
| 0.1         | 25                                        |
| 0.4         | 50                                        |
| 0.2         | 75                                        |
| 0.1         | 100                                       |

3) Find the reserves and producing life for a well whose production history is given in the following table. Exponential decline is suspected. The economic limit is 3000 bbl/yr. Use a graphical solution.

| Year | Production (bbl) X10 <sup>4</sup> |
|------|-----------------------------------|
| 1    | 20.0                              |
| 2    | 12.5                              |
| 3    | 8.0                               |
| 4    | 4.8                               |

4) Complete the evaluation of the production from the well in Q3 above by calculating the PV, assuming a stable oil price of \$50/bbl. The asset owner has an earning power of 10%.

## PNG 513 (2019/2020 1st sem.)

Pg ①

Petroleum Property Evaluation refers to the analytical procedures or economic ~~procedures~~ methods employed to assess the commercial value of oil and gas assets.

Why Petroleum Property Evaluation is Essential

- Purchasing of a producing property
- Sale of a producing property
- Collateralization of bank loans
- Basis for a decision to invest in a project
- Choosing among alternatives
- Government reporting
  - Security Exchange Commission (SEC)
  - Department of Petroleum Resources (DPR)
- To determine the value of the property.

b. Investment or cost,  $C = \$10,000$

Salvage value,  $S_v = 0$

$n = 5$  years.

$$\text{Depreciation} = \frac{C - S_v}{n}$$

$$= \frac{10000 - 0}{5} = 2000$$

| Year | Revenue (\$) | Operating Costs (\$) | Depreciation (\$) | Financial Net Income (\$) |
|------|--------------|----------------------|-------------------|---------------------------|
| 1.   | 15000        | 5000                 | 2000              | 8000                      |
| 2.   | 14000        | 6000                 | 2000              | 6000                      |
| 3.   | 13000        | 7000                 | 2000              | 4000                      |
| 4.   | 12000        | 8000                 | 2000              | 2000                      |
| 5.   | 11000        | 9000                 | 2000              | 0                         |
|      |              |                      |                   | 20,000                    |

Financial Net Income = Revenue - Operating Costs - Depreciation

Cash Flow Model

Net Cash Flow,  $NCF = \text{Revenue} - \text{Operating Costs} - \text{Investment}$

Earning power,  $i = 10\% = 0.10$

| Year | Revenue (\$) | Operating Costs (\$) | Investment Inv (\$) | NCF (\$) | $(i+1)^j$ | $\frac{NCF}{(i+1)^j}$ | $\frac{inv}{(i+1)^j}$ |
|------|--------------|----------------------|---------------------|----------|-----------|-----------------------|-----------------------|
| 0    | —            | —                    | 10000               | -10000   | 1.000     | -10000                | 10000                 |
| 1    | 15000        | 5000                 | 0                   | 10000    | 1.1000    | 9090.91               | 0                     |
| 2    | 14000        | 6000                 | 0                   | 8000     | 1.2100    | 6611.57               | 0                     |
| 3    | 13000        | 7000                 | 0                   | 6000     | 1.3310    | 4507.89               | 0                     |
| 4    | 12000        | 8000                 | 0                   | 4000     | 1.4641    | 2732.05               | 0                     |
| 5    | 11000        | 9000                 | 0                   | 2000     | 1.6105    | 1241.85               | 0                     |
|      |              |                      |                     | 20,000   |           | 14184.27              | 10000                 |

$$\begin{aligned} \text{Undiscounted Profit to Investment Ratio} &= \frac{NCF}{\text{Investment}} \\ &= \frac{20000}{10000} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{iii Discounted Profit to Investment Ratio} &= \frac{\sum \frac{NCF}{(i+1)^j}}{\sum \frac{inv}{(i+1)^j}} \\ &= \frac{14184.27}{10000} \\ &= 1.418427 \\ &= 1.42 \end{aligned}$$

Parameters:

(2)

Pg. 3

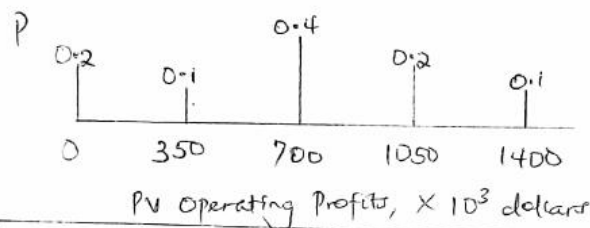
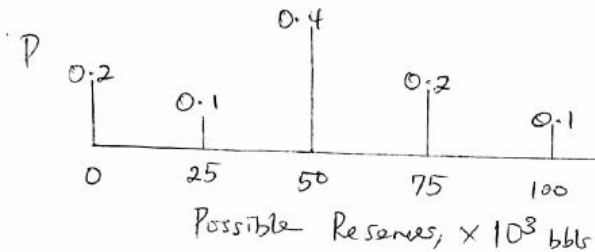
Completion cost = \$200,000

Risk = 5%

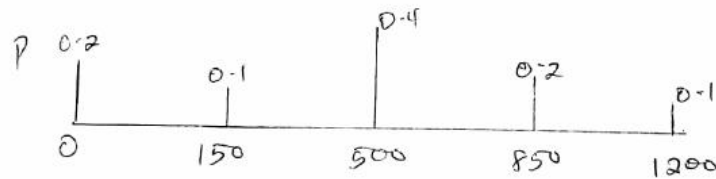
Cost per barrel = \$20/bbl

VC = Venture Capital = \$2,000,000

PV factor = 0.7



$$\text{PV Operating Profit} = (\text{Possible Reserve} \times \text{cost per barrel}) \times \text{PV factor}$$

PV Operating Profits less completion cost, x 10<sup>3</sup> dollars.

| Reserve<br>x 10 <sup>3</sup> (bbl) | P <sub>i</sub> | Operating<br>Profit (\$) x 10 <sup>3</sup> | PV of<br>Operating<br>Profit (\$) x 10 <sup>3</sup> | X <sub>i</sub> = PV of<br>operating profit<br>less completion cost | P <sub>i</sub> X <sub>i</sub> | P <sub>i</sub> (X <sub>i</sub> - $\bar{X}_A$ ) <sup>2</sup><br>X 10 <sup>6</sup> |
|------------------------------------|----------------|--------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------|
| 0                                  | 0.2            | 0                                          | 0                                                   | 0                                                                  | 0                             | 51005                                                                            |
| 25                                 | 0.1            | 500                                        | 350                                                 | 150                                                                | 15000                         | 12602.5                                                                          |
| 50                                 | 0.4            | 1000                                       | 700                                                 | 500                                                                | 200000                        | 10                                                                               |
| 75                                 | 0.2            | 1500                                       | 1050                                                | 850                                                                | 170000                        | 23805                                                                            |
| 100                                | 0.1            | 2000                                       | 1400                                                | 1200                                                               | 120000                        | 48302.5                                                                          |
|                                    |                |                                            |                                                     |                                                                    | $\bar{X}_A = 505000$          | $135725 \times 10^6$                                                             |

$$\bar{x}_A = \$505,000$$

$$\sigma = \sqrt{\sum P_i (x_i - \bar{x}_A)^2}$$

$$= \sqrt{135725 \times 10^6}$$

$$\sigma = \$368408.7404$$

$z = -1.645$  for 5% risk  $\Rightarrow$  from Monte Carlo Chart

$$w = 0, f = 1$$

$$B_1 = \frac{\bar{x}_A}{1+w} = \frac{505000}{1+0}$$

$$B_1 = \$505,000$$

$$B_2 = \frac{f}{2M} \left( \frac{z\sigma}{1+w} \right)^2$$

$$= \frac{1}{2 \times 12000000} \left( \frac{-1.645 \times 368408.7404}{1+0} \right)^2$$

$$B_2 = \$15303.13513$$

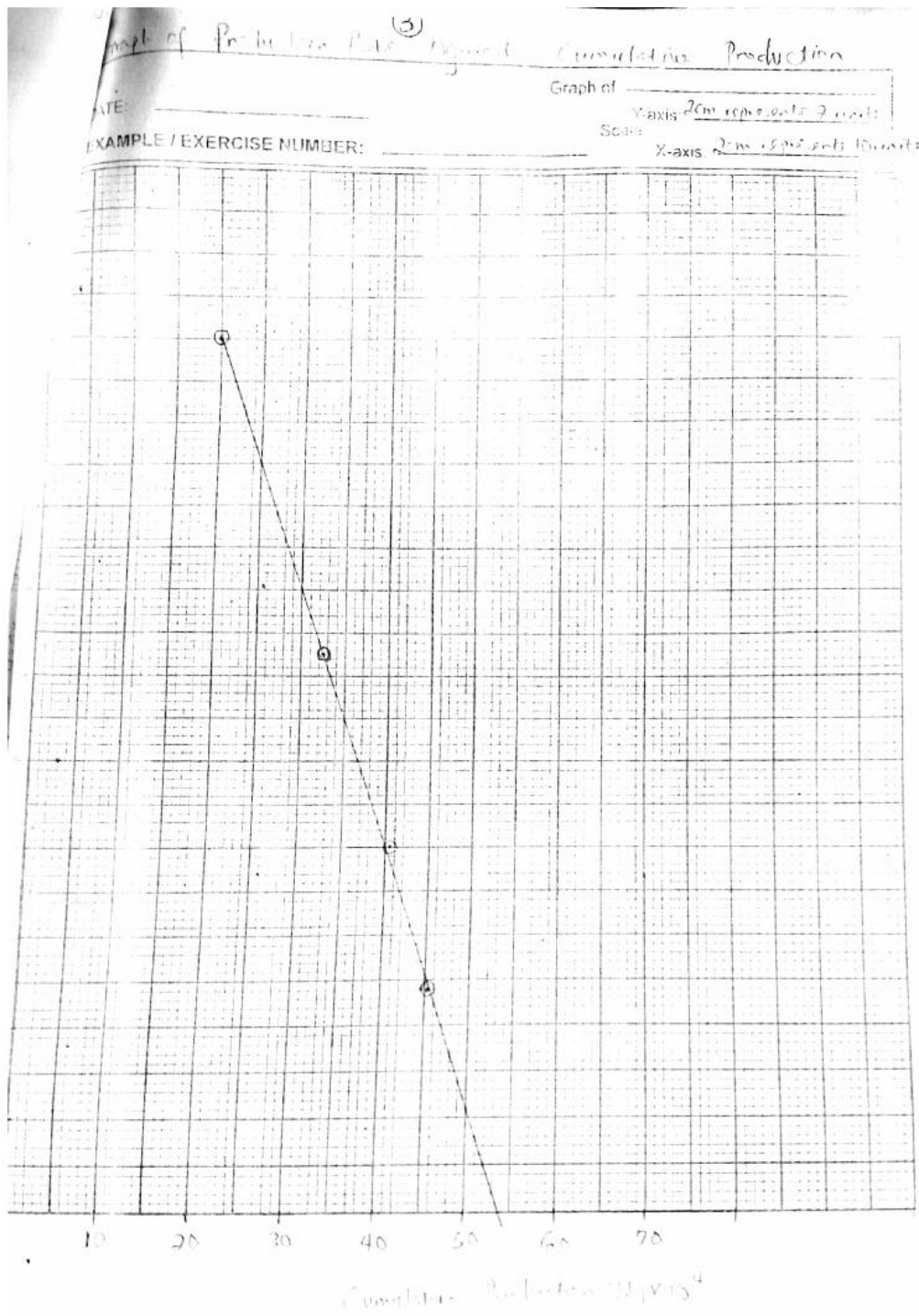
$$C \leq (B_1 + B_2) \left[ 1 - \sqrt{1 - \left( \frac{B_1}{B_1 + B_2} \right)^2} \right]$$

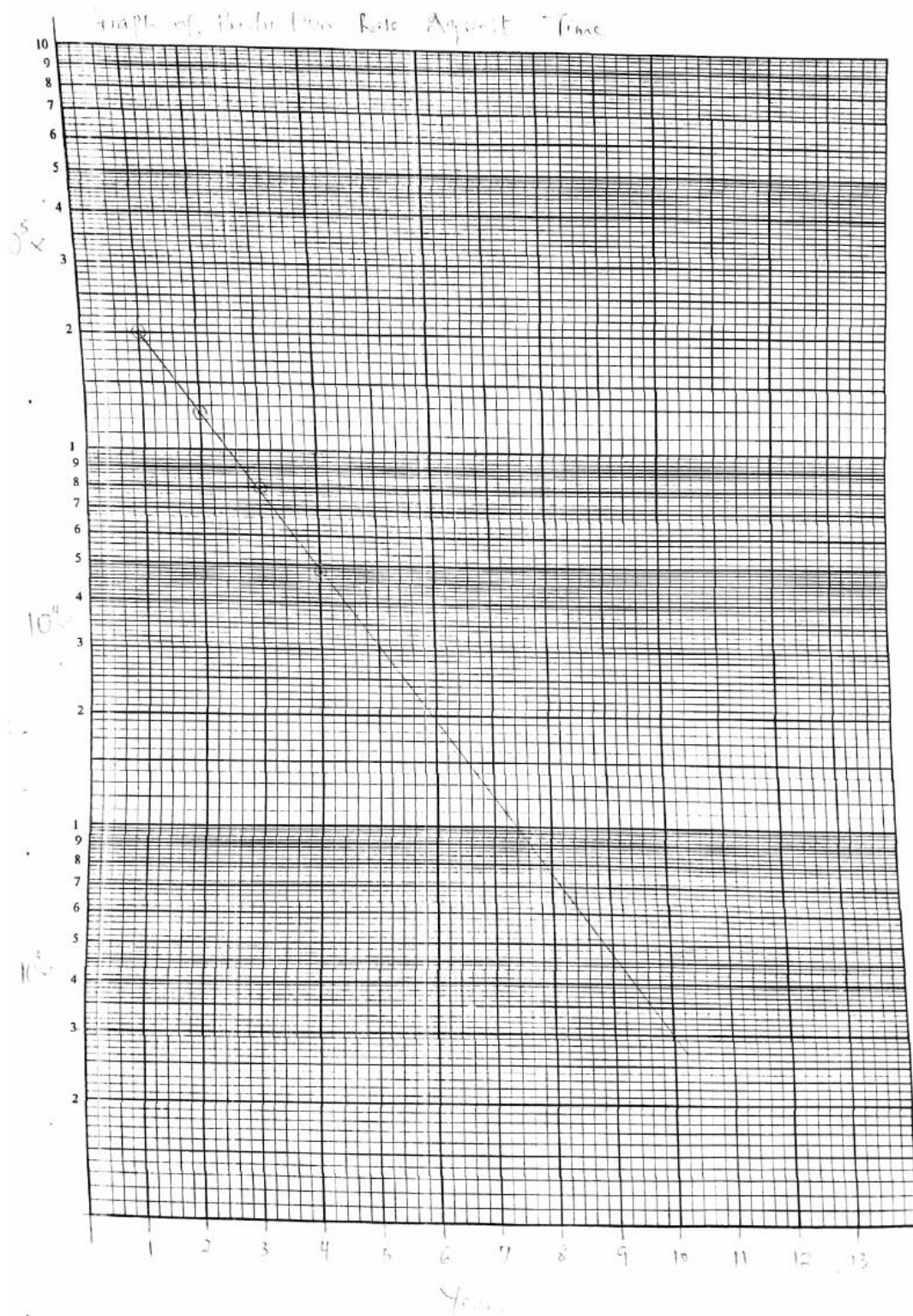
$$C \leq (505000 + 15303.13513) \left[ 1 - \sqrt{1 - \left( \frac{505000}{505000 + 15303.13513} \right)^2} \right]$$

$$C \leq 395041.9982$$

$$C \leq \$395,042$$

The maximum amount that the management should pay for the drilling cost and cost of lease is \$395,042





(3)

| Year | Production (bbl) $\times 10^4$ | Cumulative Production (bbl) $\times 10^4$ |
|------|--------------------------------|-------------------------------------------|
| 1    | 20.0                           | 20.0                                      |
| 2    | 12.5                           | 32.5                                      |
| 3    | 8.0                            | 40.5                                      |
| 4    | 4.8                            | 45.3                                      |

Pg. 5

From the Graph of Production Rate against Cumulative production  
 Ultimate Recovery =  $53.8 \times 10^4$  bbls

$$\text{Remaining Reserve} = (53.8 - 45.3) \times 10^4$$

$$= 8.5 \times 10^4$$

$$\text{Remaining Reserve} = 85,000 \text{ bbls}$$

| Year | Production (bbl)   |
|------|--------------------|
| 1    | $2.0 \times 10^5$  |
| 2    | $1.25 \times 10^5$ |
| 3    | $8.0 \times 10^4$  |
| 4    | $4.8 \times 10^4$  |

$$\text{Economic limit} = 3000 \text{ bbl/yr}$$

$$= 3 \times 10^3 \text{ bbl/yr}$$

From the graph of production rate against ~~cumulative~~ time,  
 At an economic limit of 3000 bbl/yr,  
 The producing life of the well is 10 years.

4

Oil price = \$50/bbl

(4)

Earning power,  $i = 10\%$  $= 10/100$  $i = 0.10$ 

Complete the corresponding production for each from the graph

| Year | Production (bbl) | Income =<br>Production $\times$ oil price | $(1+i)^j$ | PV = $\frac{\text{Income}}{(1+i)^j}$ |
|------|------------------|-------------------------------------------|-----------|--------------------------------------|
| 1    | 200000           | 10,000,000                                | 1.1       | 9090909.09                           |
| 2    | 125000           | 6,250,000                                 | 1.21      | 5165289.26                           |
| 3    | 80000            | 4,000,000                                 | 1.331     | 3005259.20                           |
| 4    | 48000            | 2,400,000                                 | 1.4641    | 1639232.29                           |
| 5    | 31000            | 1,550,000                                 | 1.6105    | 962434.03                            |
| 6    | 19500            | 975,000                                   | 1.7716    | 550349.97                            |
| 7    | 12000            | 600,000                                   | 1.9487    | 307897.57                            |
| 8    | 7600             | 380,000                                   | 2.1436    | 177271.88                            |
| 9    | 4700             | 235,000                                   | 2.3579    | 99664.96                             |
| 10   | 3000             | 150,000                                   | 2.5937    | 57832.44                             |
|      |                  |                                           |           | \$2,105,6140.69                      |

*Kalu's lib*  
**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN**  
**COLLEGE OF TECHNOLOGY**  
**DEPARTMENT OF PETROLEUM ENGINEERING**  
**2018/2019 First Semester Examination**  
**PNG 513: Petroleum Property Evaluation**

**Answer All Questions**

**Time: 2 Hours**

**Please write your Matriculation Number Boldly on your Script**

1) What is Petroleum Property Evaluation? Why is it a necessity in the Petroleum Industry?

b) The installation of a compressor in a gas field will generate the cash flow shown below. The yearly federal Income tax and direct costs are as shown. The required investment at the beginning of the project is \$100,000. The earning power of the asset owner is 8%.

- i) Determine the Profit to Investment Ratio using the Financial model applying unit of production depreciation.
- ii) Determine the Undiscounted Profit to Investment Ratio using the Cash Flow Model.
- iii) Determine the Discounted Profit to Investment Ratio.

| Year | Gross Income (\$) | Direct Costs (\$) | Fed Income Tax (\$) |
|------|-------------------|-------------------|---------------------|
| 1    | 100,000           | 20,000            | 17,000              |
| 2    | 90,000            | 20,000            | 20,000              |
| 3    | 80,000            | 20,000            | 16,000              |
| 4    | 60,000            | 20,000            | 4,000               |

2a) What are mutually exclusive projects? Explain how management takes a decision on them?

2b) You have a choice between the following two mutually exclusive projects with cash flows shown below.

| Year | Project A (\$) | Project B (\$) |
|------|----------------|----------------|
| 0    | -25,000        | -25,000        |
| 1    | 30,000         | 0              |
| 2    | 20,000         | 0              |
| 3    | 0              | 0              |
| 4    | 0              | 50,000         |
| 5    | 0              | 30,000         |

- i) Plot the present value profile for each of the projects on the same graph
- ii) Which project would you recommend if your reinvestment opportunity is 15%. What is the NPV at this rate?
- iii) Which project would you recommend if your reinvestment opportunity is 20%. What is the NPV at this rate?
- iv) What is the DCFROR for each project?

3) An exploratory well is giving the following payoff table:

| Probability | Present Value of Operating Profit (\$) |
|-------------|----------------------------------------|
| 0.60        | (Dry)                                  |
| 0.10        | 6,000,000                              |
| 0.20        | 12,000,000                             |
| 0.10        | 18,000,000                             |

If the well is a producer, completion costs will amount to \$2,500,000. What is the maximum dry hole cost that can be tolerated based on 5% risk? The management group has \$100MM venture capital.

4a) Name five methods for oil property (reserve) evaluation.

4b) Find the reserves and production life for a well whose production history is given in the following table. Harmonic decline is suspected. The economic limit is 1000 bbl/yr. Use a graphical solution.

| Year | Production (bbl) |
|------|------------------|
| 2016 | 90,000           |
| 2017 | 31,500           |
| 2018 | 17,500           |
| 2019 | 11,500           |

*Handwritten calculations:*  
 396404  
 27778.31  
 188276  
 116498

Petroleum Property evaluation refers to the analytical procedure by which the Commercial value of oil and gas field is assessed.

### Necessity of Petroleum Evaluation / Petroleum property Evaluation in the Petroleum Industry

- i) It is the basis for a decision to invest in a project
- ii) For Collateralization of bank loans.
- iii) For Choosing among alternatives
- iv) For sale of a producing property
- v) For purchase of a producing property.

b) Soln

$$\text{Investment} = \$100,000 \quad i = 0.18$$

- i) Using the financial model by applying unit of production depreciation (UOP)

$$\text{UOP} = \frac{\text{Income in that period} \times \text{Investment}}{\text{Total Income}}$$

$$\text{for yr 1 eg} = \frac{100,000}{330,000} \times 100,000 = 30,303.03\$$$

| Yr | Gross Income \$ | Direct Cost \$ | UOP depreciation \$ | Income taxes \$ | Profit: financial net Income Taxes \$ |
|----|-----------------|----------------|---------------------|-----------------|---------------------------------------|
| 1  | 100,000         | 20,000         | 30,303.03           | 17,000          | 32,696.97                             |
| 2  | 90,000          | 20,000         | 27,272.73           | 20,000          | 22,727.27                             |
| 3  | 80,000          | 20,000         | 24,242.42           | 16,000          | 19,757.58                             |
| 4  | 60,000          | 20,000         | 18,181.82           | 40,000          | 17,818.18                             |
|    | 330,000         |                |                     |                 | 93,000                                |

$$\text{ii) Profit to investment} = \frac{93,000 \$}{100,000 \$} = \frac{\text{Profit}}{\text{Investment}} = \underline{\underline{0.93}}$$

(1)

using Cash flow model

| Yr | Revenue<br>\$ | Direct<br>Cost \$ | Depreciation<br>Tax \$ | Investment<br>\$ | Profit<br>NCF | PV<br>NCF<br>(1+i) <sup>t</sup>        |
|----|---------------|-------------------|------------------------|------------------|---------------|----------------------------------------|
| 0  | —             | —                 | —                      | 100,000          | -100,000      | -100,000                               |
| 1  | 100,000       | 20,000            | 17,000                 | —                | 63,000        | 58,333.33                              |
| 2  | 90,000        | 20,000            | 20,000                 | —                | 50,000        | 42,866.94                              |
| 3  | 80,000        | 20,000            | 16,000                 | —                | 44,000        | 34,928.62                              |
| 4  | 60,000        | 20,000            | 4,000                  | —                | 36,000        | 26,461.08                              |
|    |               |                   |                        |                  | 93,000        | $\sum \frac{NCF}{(1+i)^t} = 62,589.97$ |

$$\text{At Yr 0} \quad \frac{NCF}{(1+i)^0} = \frac{-100,000}{(1+0.08)^0} = -100,000$$

$$\text{" " 1} \quad = \frac{63,000}{(1+0.08)^1} = 58,333.33$$

$$\text{At Yr 2} \quad = \frac{50,000}{(1+0.08)^2} = 42,866.94$$

$$\text{At Yr 3} \quad = \frac{44,000}{(1+0.08)^3} = 34,928.62$$

$$\text{At Yr 4} \quad = \frac{36,000}{(1+0.08)^4} = 26,461.08$$

⇒ undiscounted

$$\text{profit to inv ratio} = \frac{\text{undiscounted profit}}{\text{investment}} = \frac{93,000 \$}{100,000 \$} = 0.93$$

$$\Rightarrow \text{Discounted profit to investment ratio} = \frac{\sum_{t=0}^n \frac{NCF}{(1+i)^t}}{\sum \frac{inv}{(1+i)^t}} = \frac{62,589.97}{100,000} = 0.63 \text{ approx}$$

2

(2)

(2)  
Mutually Exclusive projects are two or more projects where the acceptance of one means the rejection of all others. only one project is thus undertaken.

Ways Management take decision on

Mutually Exclusive projects

(i) All the projects first pass through the Single Go or NoGo decision criteria

(ii) Each project that passes the Go or NoGo Criteria is compared to each of the remaining project.

2b) Solu

firstly find the NPV of both project at different discount rate (choose the discount rate)

we find the NPV for project A

" the NPV for project B

⇒ So find the NPV for project A at 0% 5% 10% 15% 20% 25%  
Also find the NPV for project B at 0% 5% 10% 15% 20% 25%

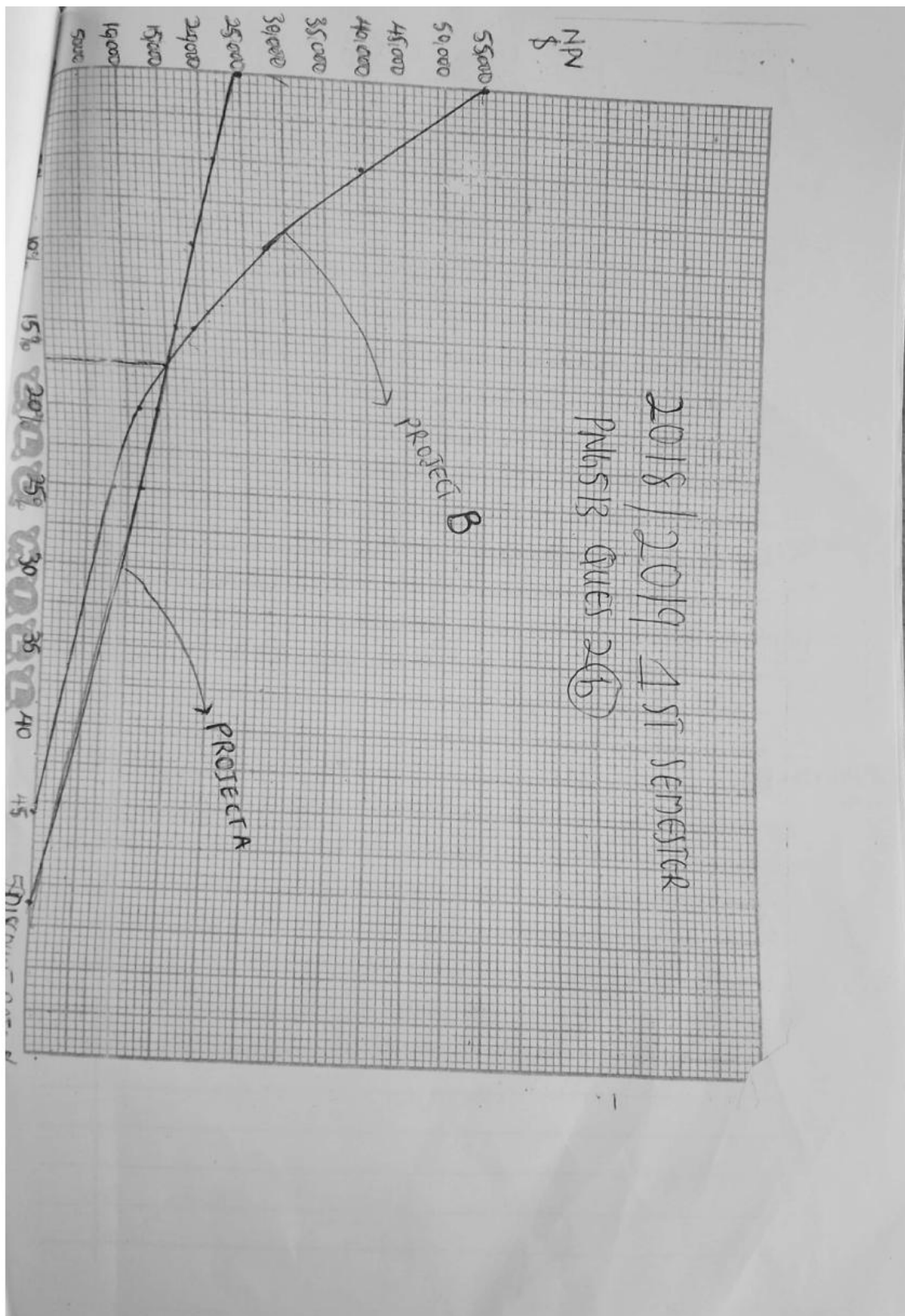
Project A

$$NPV @ 0\% = \sum PV @ 0\% \quad PV = \frac{NCF}{(1+i)^t}$$

| ③ 0% |         | PV                                  |                                         |
|------|---------|-------------------------------------|-----------------------------------------|
| Yr   | NCF     | PV @ 0%                             | PV @ 5%                                 |
| 0    | -25,000 | $\frac{-25,000}{(1+0)^0} = -25,000$ | $\frac{-25,000}{(1+0.05)^0} = -25,000$  |
| 1    | 30,000  | $\frac{30,000}{(1+0)^1} = 30,000$   | $\frac{30,000}{(1+0.05)^1} = 28,571.43$ |
| 2    | 20,000  | $\frac{20,000}{(1+0)^2} = 20,000$   | $\frac{20,000}{(1+0.05)^2} = 18,140.59$ |
| 3    | 0       | $\frac{0}{(1+0)^3} = 0$             | $\frac{0}{(1+0.05)^3} = 0$              |
| 4    | 0       | $\frac{0}{(1+0)^4} = 0$             | $\frac{0}{(1+0.05)^4} = 0$              |
| 5    | 0       | $\frac{0}{(1+0)^5} = 0$             | $\frac{0}{(1+0.05)^5} = 0$              |
|      |         | NPV = 25,000                        | NPV = 21,712.0                          |

| Yr | NCF     | PV @ 10%                               | PV @ 15%                                |
|----|---------|----------------------------------------|-----------------------------------------|
| 0  | -25,000 | $\frac{-25,000}{(1+0.1)^0} = -25,000$  | $\frac{-25,000}{(1+0.15)^0} = -25,000$  |
| 1  | 30,000  | $\frac{30,000}{(1+0.1)^1} = 27,272.73$ | $\frac{30,000}{(1+0.15)^1} = 26,086.96$ |
| 2  | 20,000  | $\frac{20,000}{(1+0.1)^2} = 16,528.93$ | $\frac{20,000}{(1+0.15)^2} = 15,122.87$ |
| 3  | 0       | 0 = 0                                  | 0                                       |
| 4  | 0       | 0 = 0                                  | 0                                       |
| 5  | 0       | 0 = 0                                  | 0                                       |
|    |         | <u>NPV = 18,801.66</u>                 | <u>NPV = 16,209.83</u>                  |

| PV @ 20%                               | PV @ 25%                               |
|----------------------------------------|----------------------------------------|
| $\frac{-25,000}{(1+0.2)^0} = -25,000$  | $\frac{-25,000}{(1+0.25)^0} = -25,000$ |
| $\frac{30,000}{(1+0.2)^1} = 25,000$    | $\frac{30,000}{(1+0.25)^1} = 24,000$   |
| $\frac{20,000}{(1+0.2)^2} = 13,888.89$ | $\frac{20,000}{(1+0.25)^2} = 12,800$   |
| 0                                      | 0                                      |
| 0                                      | 0                                      |
| 0                                      | 0                                      |
| <u>NPV = 13,888.89</u>                 | <u>NPV = 11,800</u>                    |



For Project B

Que 2 Cont'd

At discount rate 0%

NPV = 55,000

| yr | CF      | PV@ 5%                                  | PV@ 10%                                | PV@ 15%                                 |
|----|---------|-----------------------------------------|----------------------------------------|-----------------------------------------|
| 0  | -25,000 | $\frac{-25,000}{(1+0.05)^0} = -25,000$  | $\frac{-25,000}{(1+0.05)^0} = -25,000$ | $\frac{-25,000}{(1+0.15)^0} = -25,000$  |
| 1  | 0       | 0                                       | 0                                      | 0                                       |
| 2  | 0       | 0                                       | 0                                      | 0                                       |
| 3  | 0       | 0                                       | 0                                      | 0                                       |
| 4  | 50,000  | $\frac{50,000}{(1+0.05)^4} = 41,135.44$ | $\frac{50,000}{(1+0.1)^4} = 34,110.67$ | $\frac{50,000}{(1+0.15)^4} = 28,587.66$ |
| 5  | 30,000  | $\frac{30,000}{(1+0.05)^5} = 23,505.78$ | $\frac{30,000}{(1+0.1)^5} = 18,627.64$ | $\frac{30,000}{(1+0.15)^5} = 14,915.30$ |

| PV@ 20%                                | PV@ 25%                                |
|----------------------------------------|----------------------------------------|
| $\frac{-25,000}{(1+0.2)^0} = -25,000$  | $\frac{-25,000}{(1+0.25)^0} = -25,000$ |
| 0                                      | 0                                      |
| 0                                      | 0                                      |
| 0                                      | 0                                      |
| $\frac{50,000}{(1+0.2)^4} = 24,112.65$ | $\frac{50,000}{(1+0.25)^4} = 20,480$   |
| $\frac{30,000}{(1+0.2)^5} = 12,056.33$ | $\frac{30,000}{(1+0.25)^5} = 9,830.4$  |

$$NPV@ 20\% = \sum PV@ 20\% = 11,168.98$$

$$NPV@ 25\% = \sum PV@ 25\% = 5310.4$$

2ii) If the reinvestment opportunity is 15% i will recommend Project with higher NPV hence I recommend Project B  
The NPV for this project is approximately \$18,700

2iii) If the re-investment opportunity is 20% i will recommend a project with higher NPV hence I recommend Project A  
The NPV at this rate is approximately \$13,900

2iv) The DCFROR for project **B** is approximately 45.5%  
The DCFROR for project A is approximately 51%

Using the General equation to solve for C

$w=0$  with PV discounted;  $m = \$100 \times 10^6$

at 5%  $Z = -1.645$

$F=1$

| Prob | PV of operating profit (\$) | PV less completion | $P_i X_i$ | $P_i (X_i - \bar{X}_A)^2$ |
|------|-----------------------------|--------------------|-----------|---------------------------|
| 0.6  | 0                           | 0                  | 0         | $8.664 \times 10^{12}$    |
| 0.1  | 6,000,000                   | 3,500,000          | 350,000   | $9 \times 10^9$           |
| 0.2  | 12,000,000                  | 9,500,000          | 1,900,000 | $6.498 \times 10^{12}$    |
| 0.1  | 18,000,000                  | 15,500,000         | 1,550,000 | $1.3689 \times 10^{13}$   |
| 1.0  |                             |                    | 3,800,000 | $2.886 \times 10^{13}$    |

$$\bar{X}_A = 3,800,000$$

$$\sigma = \sqrt{2.886 \times 10^{13}}$$

$$\sigma = 5,372,150.407$$

From General equation is  $B_1 = \frac{X_A}{1+w} = \frac{3,800,000}{1+0} = \$3,800,000$

$$B_2 = \frac{F}{2m} \left( \frac{\sigma}{1+w} \right)^2 = \frac{1}{2 \times 100 \times 10^6} \left( \frac{-1.645 \times 3,800,000}{1+0} \right)^2$$

$$B_2 = 195,375.005$$

$$C \leq (B_1 + B_2) \left[ 1 - \sqrt{1 - \left( \frac{B_1}{B_1 + B_2} \right)^2} \right]$$

$$\left( \frac{B_1}{B_1 + B_2} \right)^2 = \left( \frac{3,800,000}{3,800,000 + 195,375.005} \right)^2$$

$$= 0.90460$$

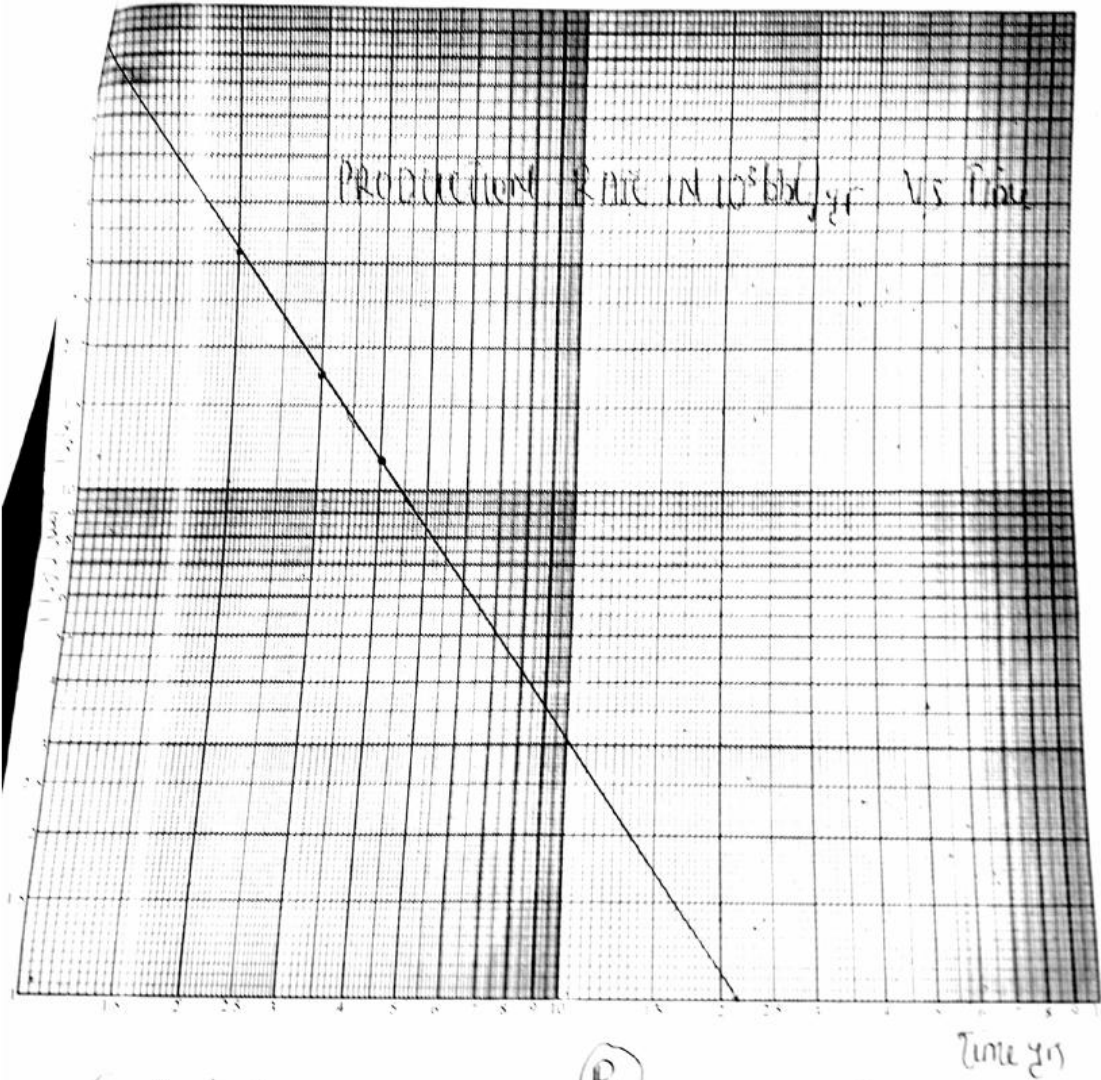
$$(B_1 + B_2) = 3,995,375.005$$

$$C \leq 3,995,375.005 \left[ 1 - 0.308885 \right]$$

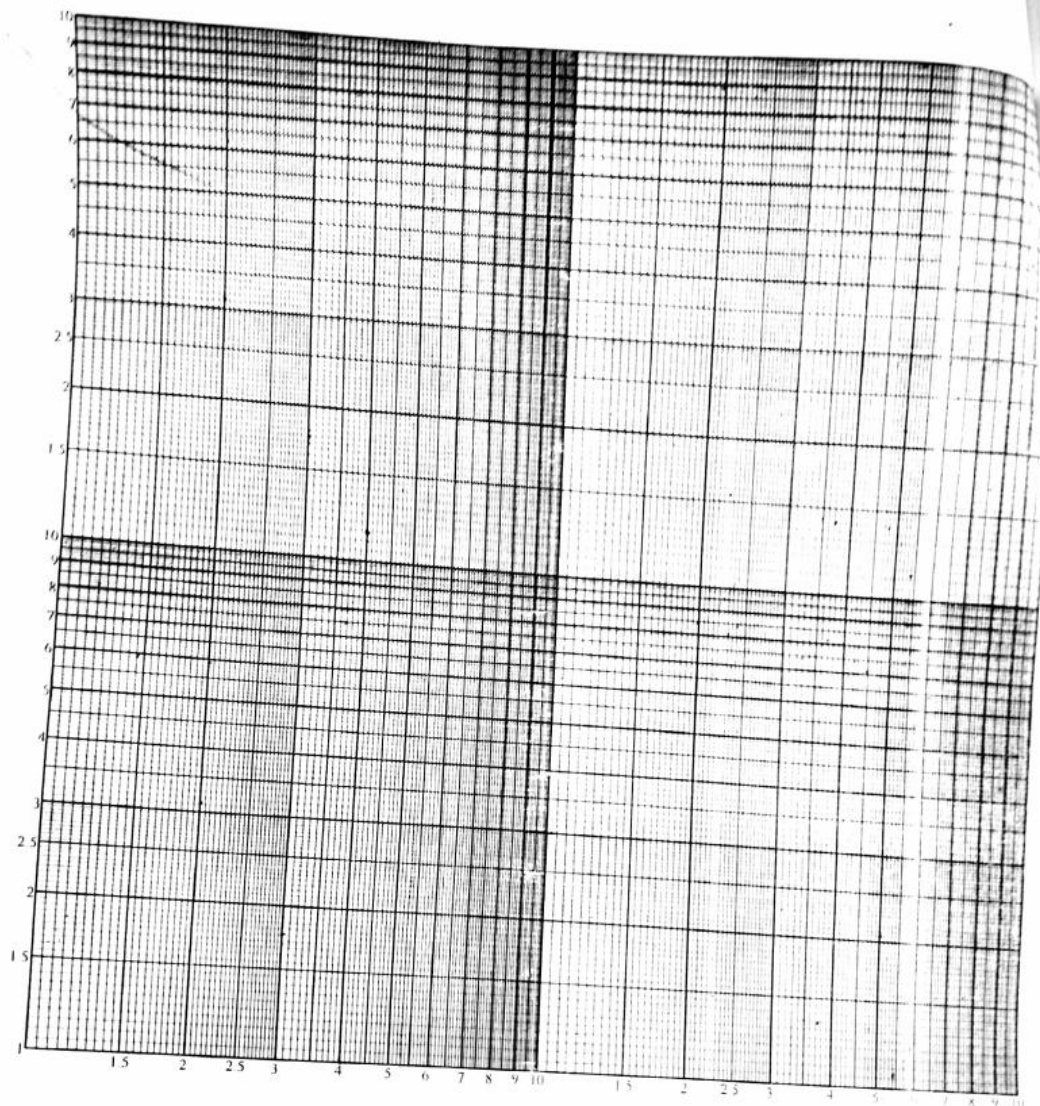
$$C \leq \$2,761,263.229$$

Management should not pay more than \$2,761,263.229

2018/2019 1st SEMESTER EXAM  
QUES 4b



(B)  
 { Production rate vs time in yrs on a log-log paper }  
 in  $10^8 \text{ bbl/yr}$



Time Years

1) 5 methods for oil property evaluation

i) Analogy

ii) Volumetrics

iii) Decline Curve

iv) Material Balance

v) Reservoir Simulation

4b) From the graph attached  
 (ultimate recovery - cumm prod)  
 Remaining reserves =  $224,000 - 150,500$   
 = 73,500 bbls

Table

| yr | Prod bbl/yr | Cumm Prod (bbl) |
|----|-------------|-----------------|
| 1  | 90,000      | 90,000          |
| 2  | 31,500      | 121,500         |
| 3  | 17,500      | 139,000         |
| 4  | 11,500      | 150,500         |

Production life from Graph B = 22 yrs

**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN**  
**COLLEGE OF TECHNOLOGY**  
**DEPARTMENT OF PETROLEUM ENGINEERING**  
**2017/2018 First Semester Examination**  
**PNG 513: Petroleum Property Evaluation**

**Answer All Questions** **Time: 2 Hours**  
**Please write your Matriculation Number Boldly on your Script**

1a) Discuss the three types of Investment Decisions that management must make and the criteria they use in each.

1b) A well is dead and needs a Coil Tubing (CT) intervention to recomplect on a new zone having appreciable reserves. This recomplect will cause the well to produce the production figures shown below. The CT well intervention will cost \$120,000. The well operating costs are \$15,000 the first year and will increase by \$1000 each of the following five years when the well will die again. The oil price is \$60/bbl and may be assumed constant. The earning power of the asset owner is 8%. The tax is included in the operating cost.

- i) Determine the Profit to Investment Ratio using the Financial model applying straight line depreciation.
- ii) Determine the Undiscounted Profit to Investment Ratio using the Cash Flow Model.
- iii) Determine the Discounted Profit to Investment Ratio.

| Year | Production (Bbls) |
|------|-------------------|
| 1    | 1000              |
| 2    | 950               |
| 3    | 900               |
| 4    | 850               |
| 5    | 800               |
| 6    | 750               |

2a) What are yardsticks. Give seven characteristics of an ideal yardstick.

2b) A company has the opportunity to participate in an offshore drilling venture. The object formation has a 50ft of pay and when productive will produce 300 bbl/acre-ft. Based on similar developments, there is estimated to be 50% chance that the venture will find no oil, a 25% chance of a field of 800 acres and a 25% chance of finding a field of 1,600 acres. The present value of the operating profit can be taken as \$2 per bbl. It will cost \$10 MM to obtain the lease, build a platform and test the structure. The company has \$30 MM invested in similar unresolved ventures. There is available an additional \$20 MM venture capital to be invested. Management is willing to assume a 5% risk. What maximum fractional participation would you propose to this company?

3a) What are Decline Curves? Discuss their requirements, advantages and disadvantages in using them for Petroleum Property Evaluation

3b) Find the reserves and production life for a well whose production history is given in the following table. Exponential decline is suspected. The economic limit is 1000 bbl/yr. Use a graphical solution.

| Year | Production (bbl) |
|------|------------------|
| 2014 | 50,000           |
| 2015 | 38,000           |
| 2016 | 28,500           |
| 2017 | 21,500           |

4) In Question 3b, determine the Present Value in 2014 Dollars of the Petroleum Resource to be produced by the well given an oil price of \$60/bbl assumed constant. The asset owner has an earning power of 10%.

3% 73/100

DEPARTMENT OF PETROLEUM ENGINEERING, FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
PNG 513 PETROLEUM ECONOMICS CLASS TEST (WRITE ON THE QUESTION PAPER) April 2018

TIME: 1 HOUR

NAME: Ogburn Success

MAT NO 1719/2013

Q1) A well is dead and needs a Coil Tubing (CT) intervention to recomplect on a new zone having appreciable reserves. This recomplect will cause the well to produce the production figures shown below. The CT well intervention will cost \$120,000. The well operating costs are \$15,000 the first year and will increase by \$1000 each of the following five years when the well will die again. The oil price is \$60/bbl and may be assumed constant. The earning power of the asset owner is 8%. The tax is included in the operating cost.

- Determine the Profit to Investment Ratio using the Financial model applying straight line depreciation.
- Determine the Undiscounted Profit to Investment Ratio using the Cash Flow Model.
- Determine the Discounted Profit to Investment Ratio.

| Year | Production (Bbls) | Year | Production (Bbls) |
|------|-------------------|------|-------------------|
| 1    | 1,000             | 4    | 850               |
| 2    | 950               | 5    | 800               |
| 3    | 900               | 6    | 750               |

| Year | Production (bbls) | Revenue \$ | Operating Cost \$ | Depreciation \$ | Financial Net Income \$ |
|------|-------------------|------------|-------------------|-----------------|-------------------------|
| 1    | 1000              | 60,000     | 15,000            | 20,000          | 25,000                  |
| 2    | 950               | 57,000     | 16,000            | 20,000          | 21,000                  |
| 3    | 900               | 54,000     | 17,000            | 20,000          | 17,000                  |
| 4    | 850               | 51,000     | 18,000            | 20,000          | 13,000                  |
| 5    | 800               | 48,000     | 19,000            | 20,000          | 9,000                   |
| 6    | 750               | 45,000     | 20,000            | 20,000          | 5,000                   |
|      |                   |            |                   |                 | \$90,000                |

$$\text{Revenue} = \text{Unit price } \$/\text{bbl} \times \text{Production}$$

$$\text{Depreciation} = \frac{C - S_v}{n} = \frac{120,000 - 0}{6} = \$20,000$$

$$\text{as } C = \$120,000$$

$$n = 6$$

$$S_v = 0$$

$$\text{Net PIR} = \frac{\text{FNI}}{\text{Investment}} = \frac{90,000}{120,000} = 0.75$$

| Year                                                               | Production<br>bbl | Revenue<br>\$ | Variable<br>Cost<br>\$ | Investment<br>\$ | NCF<br>\$ | NPV \$      |
|--------------------------------------------------------------------|-------------------|---------------|------------------------|------------------|-----------|-------------|
| 0                                                                  | —                 | —             | —                      | 120,000          | -120,000  | -120,000    |
| 1                                                                  | 1000              | 60,000        | 15,000                 | —                | 45,000    | 41,666.67   |
| 2                                                                  | 950               | 57,000        | 16,000                 | —                | 41,000    | 35,150.89   |
| 3                                                                  | 900               | 54,000        | 17,000                 | —                | 37,000    | 29,371.79   |
| 4                                                                  | 850               | 51,000        | 18,000                 | —                | 33,000    | 24,255.99   |
| 5                                                                  | 800               | 48,000        | 19,000                 | —                | 29,000    | 19,736.91   |
| 6                                                                  | 750               | 45,000        | 20,000                 | —                | 25,000    | 15,754.24   |
| Undiscounted Profit to Investment ratio = $\frac{NCF}{Investment}$ |                   |               |                        |                  | \$90,000  | \$45,936.49 |

$$= \frac{90,000}{120,000} = 0.75$$

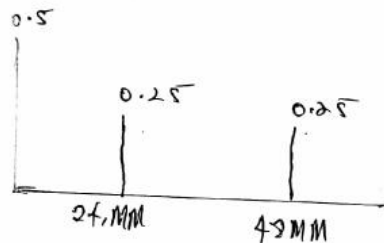
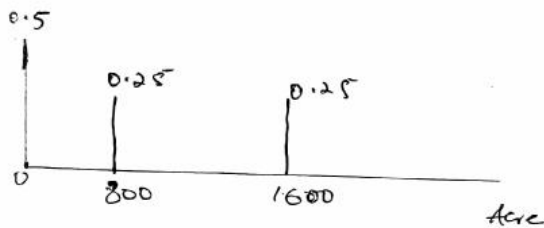
$$PV = \frac{NCF}{(1+i)^n} \quad \begin{matrix} i = 8\% \\ i = 0.08 \end{matrix}$$

| NCF \$      | NPV \$                                  |
|-------------|-----------------------------------------|
| 0. -120,000 | -120,000                                |
| 1. 45,000   | $\frac{45,000}{(1.08)^1} = \$41,666.67$ |
| 2. 41,000   | $\frac{41,000}{(1.08)^2} = \$35,150.89$ |
| 3. 37,000   | $\frac{37,000}{(1.08)^3} = \$29,371.79$ |
| 4. 33,000   | $\frac{33,000}{(1.08)^4} = \$24,255.99$ |
| 5. 29,000   | $\frac{29,000}{(1.08)^5} = \$19,736.91$ |
| 6. 25,000   | $\frac{25,000}{(1.08)^6} = \$15,754.24$ |

$$DPIR = \frac{NPV}{Investment}$$

$$= \frac{45,936.49}{120,000} = 0.3828$$

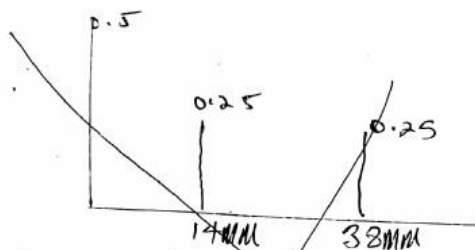
Q2) A company has the opportunity to participate in an offshore drilling venture. The object formation has a 50ft of pay and when productive will produce 300 bbl/acre-ft. Based on similar developments, there is estimated to be 50% chance that the venture will find no oil, a 25% chance of a field of 800 acres and a 25% chance of finding a field of 1,600 acres. The present value of the operating profit can be taken as \$2 per bbl. It will cost \$10 MM to obtain the lease, build a platform and test the structure. The company has \$30 MM invested in similar unresolved ventures. There is available an additional \$20 MM venture capital to be invested. Management is willing to assume a 5% risk. What maximum fractional participation would you propose to this company?



Value of operating profit

$$800 \times 2 \times 50 \times 300 = \$24MM$$

$$1600 \times 2 \times 50 \times 300 = \$48MM$$



Net operating profit less of ~~unresolved venture~~ cost to resolve lease

$$24MM - 10MM = 14MM$$

$$48MM - 10MM = 38MM$$

Correction of 1102.

| $P_i$ | Production<br>bbl | NPV of<br>operating profit<br>\$ $X_i$ | $P_i X_i$ \$ | $P_i (X_i - \bar{X}_*)^2$ |
|-------|-------------------|----------------------------------------|--------------|---------------------------|
| 0.5   | —                 | —                                      | 0            | $162 \times 10^{12}$      |
| 0.25  | 800               | 24,000,000                             | 6,000,000    | $9 \times 10^{12}$        |
| 0.25  | 1600              | 48,000,000                             | 12,000,000   | $225 \times 10^{12}$      |

$$\bar{X}_* = \sum P_i X_i = \$18,000,000.$$

$$\sigma^2 = \sum P_i (X_i - \bar{X}_*)^2 = \$396 \times 10^{12}$$

$$\therefore \sigma = \$19899748.74$$

$$\therefore M = \$30MM + \$20MM = \$50MM.$$

$$C = \$10MM.$$

$$\text{at } 5\% \text{ risk } Z = -1.645.$$

$$F \leq \frac{M}{C} \left[ \frac{\bar{X}_* - C(1+w)}{Z\sigma} \right]^2$$

$$F \leq \frac{50 \times 10^6}{10 \times 10^6} \left[ \frac{18 \times 10^6 - 10 \times 10^6(1+0)}{-1.645 \times 19899748.74} \right]^2$$

$$\therefore F \leq 0.2986.$$

$$\therefore F \leq 29.86\%.$$

Hence, the Company should not take More than 29.86% in at 5% risk.

3.

| year | Prod. (bbl) | Cumm. Prod. (bbl) |
|------|-------------|-------------------|
| 2014 | 50,000      | 50,000            |
| 2015 | 38,000      | 88,000            |
| 2016 | 28,500      | 116,500           |
| 2017 | 21,500      | 138,000           |

For exponential decline:

- to plot a graph of Prod vs time on a Semi-log plot to obtain the life of the well
- to plot the graph of Prod vs Cumm. Prod. on a Cartesian graph to obtain the reserve.

$z = 5$   
 $\lambda = 5$   
 $\sigma = 5$   
 $\sigma = 5$

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**DEPARTMENT OF PETROLEUM & NATURAL GAS ENGINEERING**  
**2015/2016 First Semester Examination**  
**PNG 513: Petroleum Property Evaluation**

**Answer All Questions** **Time: 2 Hours**  
**Please write your Matriculation Number Boldly on your Script**

- 1a) Discuss briefly why oil property evaluation is important.
- 1b) The reservoir pressure of a Niger Delta field has declined to the point that Well ND-1 can no longer produce naturally. As a result, it has been decided that a downhole electric pump having a service life of six years will be purchased for \$12,000. The pump will cause the well to produce the oil production data shown in the table below.

| Year | Production (bbls) |
|------|-------------------|
| 1    | 1000              |
| 2    | 950               |
| 3    | 900               |
| 4    | 850               |
| 5    | 800               |
| 6    | 750               |

Operating costs are estimated to be \$7,500 the first year and will increase by \$500 each of the following five years. The pump has no salvage value. The yearly tax rate for the project is 46% of PBIT. Assume a constant oil price of \$20/bbl.

- Calculate the financial net income AFIT of the project using UOP depreciation
- Carry out a NCF analysis of the project assuming the yearly taxes calculated before remain the same

2a) A National Oil Company engages in oil exploration to increase in stature. Explain how this is achieved.

2b) An exploratory well has the prospects given in the table below. Completion costs require an additional \$30,000. Management has \$1 MM venture capital and will accept only 5% risk. How much can be paid for a dry well?

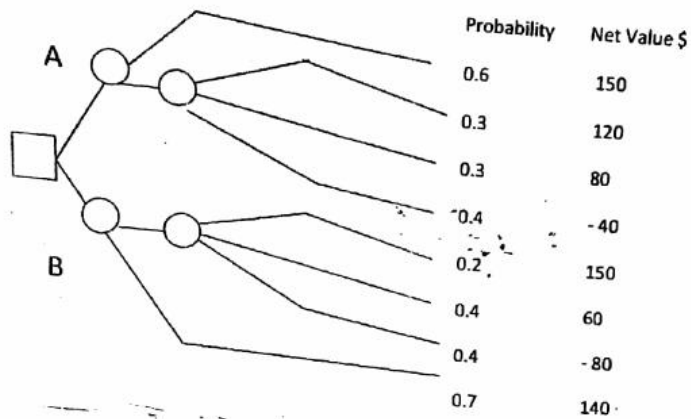
| Probability | Present Value of Operating Profits (\$) |
|-------------|-----------------------------------------|
| 0.65        | (dry)                                   |
| 0.10        | 50,000                                  |
| 0.10        | 100,000                                 |
| 0.10        | 150,000                                 |
| 0.05        | 200,000                                 |

*max profit  
no loss of profits*

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3a) What are Decision Trees and Probability Trees?

3b) Calculate the mean, variance and standard deviation of each branch that leaves the decision node in the decision tree below. Branches that have lower expected value and greater variance are often trimmed from decision trees. Simply the tree on this basis. Which branch should be trimmed?



4a) Explain how the complete evaluation of a petroleum reservoir involves the march of time.

4b) Find the reserves and production life for a well whose production history is given in the following table. Exponential decline is suspected. The economic limit is 1000 bbl/yr. Use a graphical solution.

| Year | Production (bbl) |
|------|------------------|
| 1    | 31,000           |
| 2    | 20,500           |
| 3    | 13,600           |
| 4    | 9,000            |
| 5    | 6,000            |

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2015/2016.

Financial Model.

| Year | Production (bbl) | Revenue \$ | Direct Cost \$ | WOP \$   | Income BT \$ | 46% tax \$ | Profit AFIT \$ |
|------|------------------|------------|----------------|----------|--------------|------------|----------------|
| 1.   | 1,000            | 20,000     | 7500           | 2285.714 | 10214        | 4698       | 5516           |
| 2.   | 950              | 19000      | 8000           | 2171.429 | 8829         | 4061       | 4768           |
| 3.   | 900              | 18000      | 8500           | 2057.143 | 7443         | 3424       | 4019           |
| 4.   | 850              | 17000      | 9000           | 1942.857 | 6057         | 2786       | 3271           |
| 5.   | 800              | 16000      | 9500           | 1828.571 | 4671         | 2149       | 2522           |
| 6.   | 750              | 15000      | 10,000         | 1714.286 | 3286         | 1512       | 1774           |
|      |                  |            |                |          |              |            | \$21876        |

Hence, FNI AFIT = \$21876

CASH FLOW MODEL

| Year | Production bbl | Revenue \$ | Direct Cost \$ | Investment \$ | Income BFIT \$ | 46% tax remain the same \$ | NCF AFIT \$ |
|------|----------------|------------|----------------|---------------|----------------|----------------------------|-------------|
| 0    | —              | —          | —              | 12000         | -12000         | —                          | -12000      |
| 1    | 1000           | 20,000     | 7500           | —             | 12500          | 4698                       | 7802        |
| 2    | 950            | 19000      | 8000           | —             | 11000          | 4061                       | 6939        |
| 3    | 900            | 18000      | 8500           | —             | 9500           | 3424                       | 6076        |
| 4    | 850            | 17000      | 9000           | —             | 8000           | 2786                       | 5214        |
| 5    | 800            | 16000      | 9500           | —             | 6500           | 2149                       | 4351        |
| 6    | 750            | 15000      | 10,000         | —             | 5000           | 1512                       | 3488        |
|      |                |            |                |               |                |                            | \$21870     |

Hence NCF AFIT = \$21870

| $P_i$ | PV of net operating profit<br>\$ | Completion cost<br>\$ | PV less completion cost<br>\$ $X_i$ | $P_i X_i$<br>\$ | $P(X_i - \bar{X}_A)^2 \times 10^6$<br>\$ |
|-------|----------------------------------|-----------------------|-------------------------------------|-----------------|------------------------------------------|
| 0.65  | 0                                | 0                     | 0                                   | 0               | 565.76                                   |
| 0.10  | 50,000                           | 20,000                | 20,000                              | 2000            | 9.03                                     |
| 0.10  | 100,000                          | 30,000                | 70,000                              | 7000            | 164.03                                   |
| 0.10  | 150,000                          | 30,000                | 120,000                             | 12000           | 219.03                                   |
| 0.05  | 200,000                          | 30,000                | 170,000                             | 8500            | 987.01                                   |
|       |                                  |                       | \$29500                             |                 | \$2544.76 \times 10^6                    |

$$\therefore \bar{X}_A = \$29500,$$

$$D^2 = \$2544.76 \times 10^6$$

$$D = \sqrt{2544.76 \times 10^6} \equiv \$50.446 \times 10^3 \\ \equiv \$50,445.6$$

$$Z = -1.645$$

$$w = 0$$

$$M = \$1,000,000$$

$$B_1 = \frac{X_A}{1+w} = \frac{29500}{1+0} = \$29500$$

$$B_2 = \frac{F}{2M} \left( \frac{ZD}{1+w} \right)^2 = \frac{1}{2 \times 10^6} \left( \frac{-1.645 \times 50445.6}{1+0} \right)^2 \\ = \$3443.09$$

$$\text{debt } C = (B_1 + B_2) \left( 1 - \sqrt{1 - \left( \frac{B_1}{B_1 + B_2} \right)^2} \right)$$

$$C = (29500 + 3443.09) \left[ 1 - \sqrt{1 - \left( \frac{29500}{29500 + 3443.09} \right)^2} \right]$$

$$\therefore C = 32943.09 \times 0.5549$$

$$\therefore C = \$18280.31$$

Conclusion: Management should not pay more than \$18280.31 for a dry well.

OR

The Maximum dry hole cost that can be tolerated is \$18280.31.

For Decision A.

| $P_i$ | NPV<br>\$ $X_i$ | $P_i X_i$<br>\$ | $P(X_i - \bar{X})^2$<br>\$ |
|-------|-----------------|-----------------|----------------------------|
| 0.6   | 150             | 90              | 1078.656                   |
| 0.12  | 120             | 14.4            | 18.4512                    |
| 0.12  | 80              | 9.6             | 91.4112                    |
| 0.16  | -40             | -6.4            | 3485.722                   |
|       |                 | \$107.6         | \$4674.24                  |

$$\bar{X} = \$107.6$$

$$\sigma = \sqrt{4674.24} = \$68.368$$

For Decision B.

| $P_i$ | NPV<br>\$ $X_i$ | $P_i X_i$<br>\$ | $P_i (X_i - \bar{X})^2$ |
|-------|-----------------|-----------------|-------------------------|
| 0.06  | 150             | 9               | 1039.1136               |
| 0.12  | 60              | 7.2             | 232.6992                |
| 0.12  | -80             | -9.6            | 1564.9968               |
| 0.7   | 140             | 98              | 30.492                  |
|       |                 | \$104.6         | \$2873.3016             |

$$\therefore \bar{X}_B = \$104.6$$

$$s = \sqrt{2873.3016} = \$53.60$$

Conclusion:

| year | Prod. (bbl) | Cumm. Prod. (bbls) |
|------|-------------|--------------------|
| 1    | 31000       | 31000              |
| 2    | 20500       | 51500              |
| 3    | 13600       | 65100              |
| 4    | 2000        | 67100              |
| 5    | 6000        | 73100              |

For exponential decline

- f. plot the graph of prod vs time on a semi-log plot to obtain the life of the well
- g. plot the graph of prod vs Cumm. prod on a Cartesian graph to obtain the reserve.

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**DEPARTMENT OF PETROLEUM & NATURAL GAS ENGINEERING**  
**2014/2015 First Semester Examination**  
**PNG 513: Petroleum Property Evaluation**

**Answer All Questions** **Time: 3 Hours**  
**Please write your Matriculation Number Boldly on your Script**

1a) What is oil property evaluation?

1b) An oil field transfer pump having a useful life of five years is purchased for \$1,000. The pump will generate an incremental \$1,500 in the first year. However the revenue will decrease \$100 per year in each of the next four years. Operating costs will be \$500 in the first year but will increase \$100 per year in each of the following four years.

- Calculate the Financial net income before income taxes for each year using straight line depreciation noting that there is no salvage value
- Prepare a net cash flow analysis for this investment.

2a) Explain the types of investment decisions that the management of a company has to deal with from time to time.

2b) Money is worth 1% per month. An Indigenous Operator approaches a bank for a loan facility to buy a compressor unit to be installed in a gas plant. The bank offers the company two deals as follows:

**Deal A**

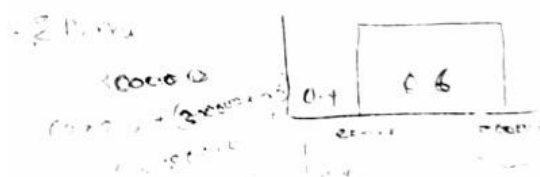
- Pay no money down.
- Pay \$800 per month for one year.
- Pay \$1000 per month for the second year.
- Pay \$1100 per month for years 3, 4, and 5

**Deal B**

- Pay \$X amount down
- Pay \$1000 per month for 5 years.

What is the amount X that the company must pay down to make Deal A and Deal B equivalent?

3) An exploratory well if successful will be completed with an additional \$50,000 cost. The PV of net operating profits can be represented by a uniform distribution with a spike. The spike of zero PV has a probability of 0.4 while the uniform distribution for a successful well lies between \$200,000 and \$500,000. Management will take only 5% risk and has a speculative budget of \$1.0 million. What is the cost of a dry well?



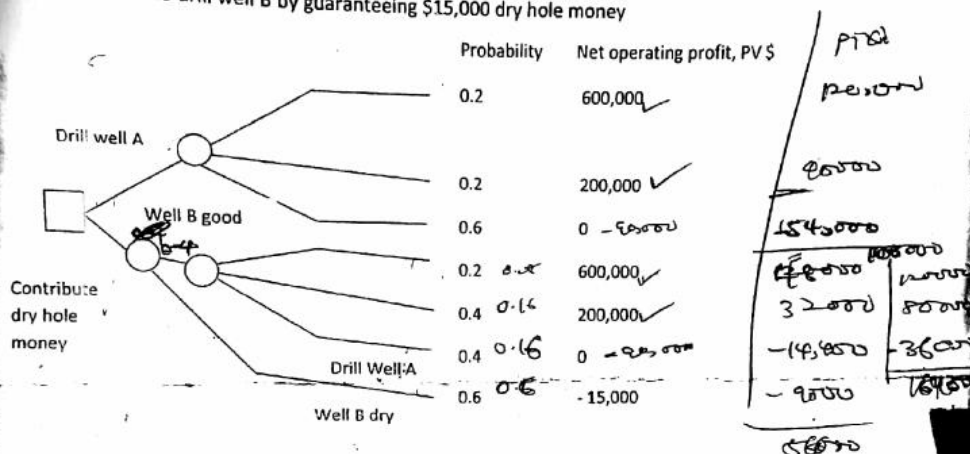
Handwritten equations:

$$\bar{R}_A = \sum P_i R_i$$

$$0.4(0) + (0.6) \left( \frac{200,000 + 500,000}{2} \right)$$

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An operator W is contemplating drilling well A on his lease. A dry hole will cost \$90,000 and a completed well will cost \$150,000. Another operator has an adjacent lease on which he may drill well B. This is contingent upon operator W contributing \$15,000 toward the expenses of well B in the event that it turns out dry. However, if well B turns out to be good, the prospects for well A would be much brighter, as indicated on the diagram. According to expected values, should operator W proceed immediately to drill well A, or should he encourage the other operator to drill well B by guaranteeing \$15,000 dry hole money



5a) What are decline curves, their benefits and constraints?

5b) Find the reserves and production life for a well whose production history is given in the following table. Harmonic decline is suspected. The economic limit is 1000 bbl/yr. Use a graphical solution.

| Year | Production (bbl) | Cum. Prod |
|------|------------------|-----------|
| 1    | 31,000           | 31,000    |
| 2    | 15,400           | 46,400    |
| 3    | 10,300           | 56,700    |
| 4    | 7,300            | 64,000    |
| 5    | 6,100            | 70,100    |

Decline curves are curves that shows how oil & gas production rates decrease over time  
 D.C.A is a means of predicting future oil well & gas well production based on past production history

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2014/2015.

Financial Model.

| Year | Revenue<br>\$ | Direct Cost<br>\$ | Depreciation<br>\$ | Net Financial Income BT.<br>\$ |
|------|---------------|-------------------|--------------------|--------------------------------|
| 1.   | 1500          | 500               | 200                | 800                            |
| 2.   | 1400          | 600               | 200                | 600                            |
| 3.   | 1300          | 700               | 200                | 400                            |
| 4.   | 1200          | 800               | 200                | 200                            |
| 5.   | 1100          | 900               | 200                | 0                              |
|      |               |                   |                    | \$2,000.                       |

$$C = \$1000.$$

$$S_v = 0.$$

$$\text{Depreciation} = \frac{C - S_v}{n}$$

$$\therefore D = \frac{1000 - 0}{5} = \$200.$$

## Cash flow Model.

| Year | Revenue<br>\$ | Direct Cost<br>\$ | Investment<br>\$ | NCFBT.           |
|------|---------------|-------------------|------------------|------------------|
| 0    | —             | —                 | 1000             | -1000.           |
| 1    | 1500          | 500               | —                | 1000             |
| 2    | 1400          | 600               | —                | 800              |
| 3    | 1300          | 700               | —                | 600              |
| 4    | 1200          | 800               | —                | 400              |
| 5    | 1100          | 900               | —                | 200.             |
|      |               |                   |                  | NS 2050.<br>NFBT |

$$3. \quad P = \frac{1100}{\left( \frac{0.01}{(0.01+1)^{36}} + 0.01 \right)} = \$33118.3$$

$$PV = \frac{33118.3}{(1+0.12)^2} = \$26401.$$

$$NPV = 9604.1 + 10049.2 + 26401 = \$45,455.$$

Def B.

$X \rightarrow$  Amount to be paid down now  $PV = \$X$ .

$$P = \frac{1000}{\left( \frac{0.01}{(1+0.01)^{60}} + 0.01 \right)}$$

$$= \$44955.04$$

$\therefore$  Def A = Def B.

$$45455 = X + 44955.04$$

$$\therefore X = \$499.7$$

$$X = \$500.$$

Deal A.

2b.

$$\text{from } A = p \left( \frac{i}{(1+i)^n - 1} + i \right)$$

$$\therefore p = \frac{A}{\left( \frac{i}{(1+i)^n - 1} + i \right)}$$

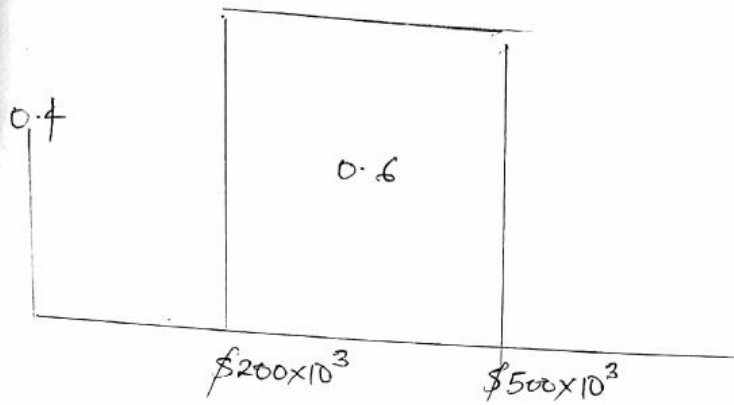
$$\text{for year 1. } p = \frac{200}{\left( \frac{0.01}{(1+0.01)^{12} - 1} + 0.01 \right)} = \$9004.1$$

$$\text{year 2, } p = \frac{1000}{\left( \frac{0.01}{(1+0.01)^{12} - 1} + 0.01 \right)} = \$11255.1$$

$$i = 0.01$$

$$i_n = i \times M = 0.01 \times 12 = 0.12$$

$$\therefore PV = \frac{p}{(1+i)^n} = \frac{11255.1}{(1+0.12)^1} = \$10049.2$$

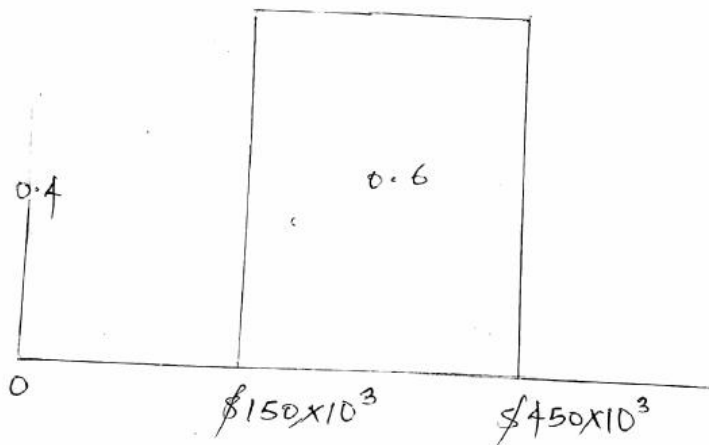


PV net operating profit.

Completion Cost = \$50,000.

Risk = 5%.

M = \$1,000,000.



PV net operating profit less Completion.

Considering the Spite probability.

Part 1:

$$M_1 = 0, \quad \sigma_1 = 0.$$

Part 2:

$$M_2 = \frac{q_2 + b_2}{2} = \frac{150 \times 10^3 + 450 \times 10^3}{2} = \$300 \times 10^3.$$

$$\sigma_2 = \frac{(b_2 - q_2)^2}{12} = \frac{(450 \times 10^3 - 150 \times 10^3)^2}{12} = \$7.5 \times 10^9.$$

$$\bar{X}_A = \sum p_i M_i$$

$$= (0.4 \times 0) + (0.6 \times 300 \times 10^3) = \$180,000.$$

$$\sigma_i^2 = \sum p_i (\sigma_i^2 + (M_i - \bar{X}_A)^2)$$

$$= 0.4 (\sigma^2 + (0 - 180 \times 10^3)^2) + 0.6 ((7.5 \times 10^9)^2 + (300 \times 10^3 - 180 \times 10^3)^2)$$

$$= \$13.796 \text{ MMM} + 13.14 \text{ MMM}$$

$$\sigma^2 = 2.61 \times 10^{10}$$

$$\sigma = \sqrt{2.61 \times 10^{10}}$$

$$\sigma = \sqrt{2.61 \times 10^{10}} = \$1.62 \times 10^5.$$

Summary

$$\bar{X}_A = \$180,000, \quad \sigma = \$162,000$$

Using the general risk equation

$$B_1 = \frac{\bar{X}_A}{1+w} = \frac{180,000}{1+0} = \$180,000$$

$$B_2 = \frac{f}{2M} \left( \frac{z\sigma}{1+w} \right)^2 = \frac{1}{2 \times 10^6} \left( \frac{-1.645 \times 162,000}{1+0} \right)^2$$

$$= \$35,508.46$$

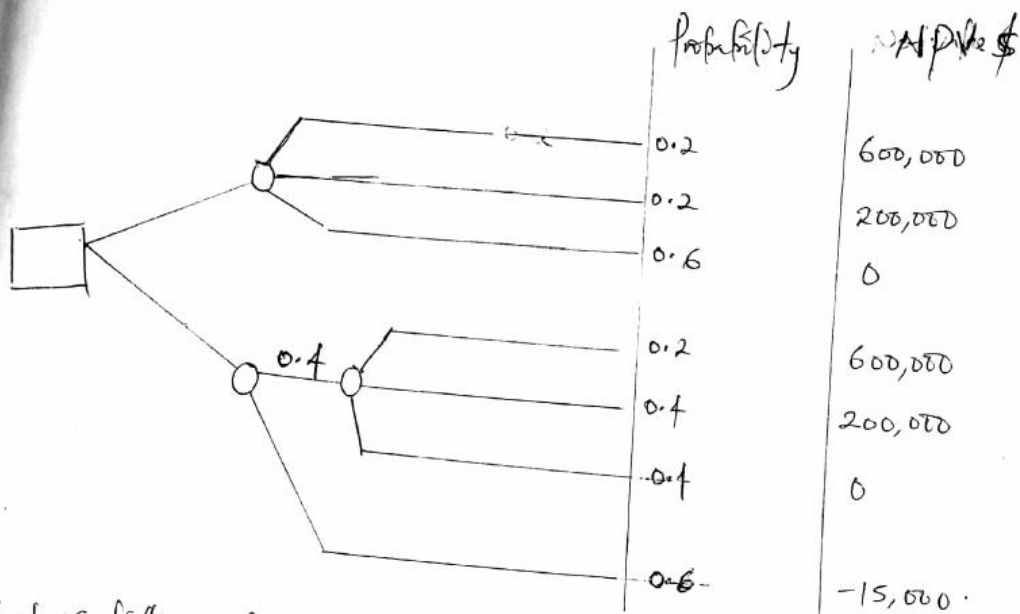
$$C = (B_1 + B_2) \left[ 1 - \sqrt{1 - \left( \frac{B_1}{B_1 + B_2} \right)^2} \right]$$

$$= (180,000 + 35,508.46) \left[ 1 - \sqrt{1 - \left( \frac{180,000}{180,000 + 35,508.46} \right)^2} \right]$$

$$= 215,508.46 \times 0.4501$$

$$\therefore C = \$97,001.51$$

Conclusion: Management should not pay More than \$97,001.51 for a dry well



Cost of drilling a dry hole = \$90,000.

Completed well will cost = \$150,000.

Recall that the cost of drilling a dry hole is inclusive in the cost of completion.

Decision A.

| $p_i$ | NPV     | drilling cost | Completion cost | Net Present Value $X_i$ | $p_i X_i$ | $p_i (X_i - \bar{X}_A)^2$ |
|-------|---------|---------------|-----------------|-------------------------|-----------|---------------------------|
| 0.2   | 600,000 | -90,000       | 60,000          | 450,000                 | 90,000    | 326432000000              |
| 0.2   | 200,000 | -90,000       | 60,000          | 50,000                  | 10,000    | 32000000                  |
| 0.6   | 0       | -90,000       | 60,000          | -90,000                 | -54,000   | 110976000000              |

$$\bar{X}_A = \$46,000$$

$$\therefore \sigma^2 = \sum p(x - \bar{x}_x)^2 = 43744000000.$$

$$\therefore \sigma = \$209150.66$$

Decision B

| $P_i$ | NPV<br>\$ | drilling cost<br>\$ | completion<br>cost \$ | Net Present<br>Value \$ $X_i$ | $P_i X_i$ | $P_i (x - \bar{x}_x)^2$ |
|-------|-----------|---------------------|-----------------------|-------------------------------|-----------|-------------------------|
| 0.16  | 600,000   | 90,000              | 60,000                | 450,000                       | 72,000    | 2598540000              |
| 0.16  | 260,000   | 90,000              | 60,000                | 50,000                        | 8,000     | 774400                  |
| 0.08  | 0         | 90,000              | —                     | -90,000                       | -7,200    | 1519107200              |
| 0.6   | -15,000   | —                   | —                     | -15,000                       | -9,000    | 2366304000              |

$$\therefore \bar{x}_x = \$77,200.$$

$$\sigma^2 = \sum p_i (x - \bar{x}_x)^2 = 29871625600.$$

$$\sigma = \$172834.099.$$

Decision

We choose a project with high Mean ( $\bar{x}_x$ ) and low Variance ( $\sigma$ ) to guarantee success.  
Hence we encourage the operator to drill well B by guaranteeing \$15,000 dry hole Money.

0609/2011

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
COLLEGE OF TECHNOLOGY

2013/2014 First Semester Examination  
PNG 513: Petroleum Property Evaluation

(4a)

Reasons

Answer Questions 1 – 4 and either Question 5 or Question 6

Time: 3 Hours

Please write your Matriculation Number Boldly on your Script

Buying

1) Sale of ✓  
producing  
property ✓

3) Collateralization  
of Bank loan

4) Government report

5) Decision making  
on project to  
invest in

6) Choosing among  
alternatives

Give five reasons why a Petroleum Property may be evaluated.

The reservoir in a Niger Delta field has declined to a point that that Well 1 will no longer flow naturally. As a result it has been decided that a down-hole electric pump having a service life of six years be purchased for \$12,000. The pump will cause the well to produce the oil production as shown below. Operating costs are estimated to be \$7,500 for the first year and will increase by \$500 each of the following five years. The pump has no salvage value. Calculate the gross revenue using a constant oil price of \$20/bbl. If the tax rate (royalty inclusive) is 46%, calculate the net income AFIT for each year using UOP depreciation. What is the profit from this investment?

| Year | Prod (bbls) |
|------|-------------|
| 1    | 1,000       |
| 2    | 950         |
| 3    | 900         |
| 4    | 850         |
| 5    | 800         |
| 6    | 750         |

DARR

2) PBT

3) Urgency

4) UPIR

5) Bail out feature

Name five characteristics of an ideal yardstick for measuring investment worth.

Give five yardsticks that ignore time value of money and five yardsticks that take time value of money into account.

A project requires the investment of \$100,000. The return cash flow will occur in equal amounts at the ends of the next 10 years. If a rate of return on investment of 20% is required, what must be the amount of each payment?

NPV

PVP

DCFROR (IFOR)

DPIR

Incremental DCFOR

An exploratory well has the prospects given in the table below. The cost of drilling a dry hole is \$150,000. Completion costs require an additional \$300,000. The management group has \$10 million venture capital and will take only 5% risk. How much can be paid for the lease?

| Probability | PV of Operating Profits (\$) |
|-------------|------------------------------|
| 0.65        | 0                            |
| 0.10        | 500,000                      |
| 0.10        | 1,000,000                    |
| 0.10        | 1,500,000                    |
| 0.05        | 2,000,000                    |

NCF =

$$\frac{23852.28}{(1+i)^n}$$

$$A = F \left[ \frac{i}{(1+i)^n - 1} \right]$$

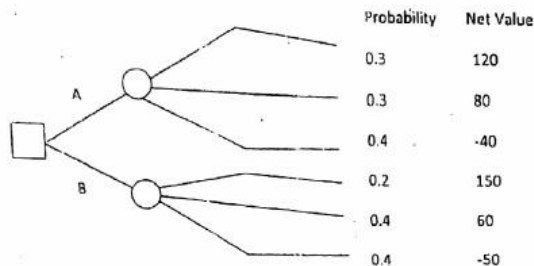
$$= \frac{0.2}{(1.2)^{10} - 1} + 0.2$$

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CS CamScanner

4a) What are Decision and Probability Trees?

4b) Calculate the mean, variance and standard deviation of each branch that leaves the Decision node. Branches that have lower expected value and greater variance are often trimmed from Decision trees. On this basis, can the tree be simplified?



220 R. 75

5) A workover is being planned in which a zone, presently behind casing will be tested. This will require a workover rig, perforating, treating and testing. Productive portions of the formation will produce 150 bbl/acre-ft. Spacing is 40 acres. There is doubt about the thickness of the possible productive interval, and the prospects have been summarized in the table. The operating profit will be \$2 per bbl and a PV factor of 0.7 is applicable. The company will engage in 20 of such projects during the current program. On this basis, what is the maximum that could be invested in the workover, so that at least a 50% present value / investment ratio could be realized, with 90% confidence?

| Probability | Net Producing Interval (ft) |
|-------------|-----------------------------|
| 0.4         | 0                           |
| 0.2         | 15                          |
| 0.2         | 20                          |
| 0.2         | 25                          |

$$\frac{600000}{(46000 - 10000) \ln \left( \frac{146000}{10000} \right)} \times 40 \times 15 \text{ acre-ft} \times 600 \text{ acre-ft}$$

6a) Give five methods for Property (Reserve) Estimation

- 1) Analog
- 2) Volumetric
- 3) Material Balance
- 4) Decline Curve
- 5) Reservoir Simulation

6b) An oil property is estimated to have a total reserves amounting to 800,000 bbl. It will produce unrestricted from an initial rate of 400 bbl/day to an economic limit of 30 bbl/day. Calculate the total time on production by assuming (i) exponential decline, (ii) general hyperbolic decline with  $b = 0.5$  and (iii) harmonic decline.

when  $b = 0$

$$t_f = \frac{NPV}{q_0 - q_f} \ln \left( \frac{q_0}{q_f} \right)$$

$$= \frac{800000}{400 - 30} \ln \left( \frac{400}{30} \right)$$

$$= 5600.6 \text{ yrs}$$

when  $b = 0.5$

$$t_f = \frac{NPV}{q_0} \sqrt{\frac{q_0}{q_f}}$$

$$= \frac{800000}{400} \sqrt{\frac{400}{30}}$$

$$= 7302.99 \text{ yrs}$$

$$= 7303 \text{ yrs}$$

when  $b = 1$

$$t_f = \frac{NPV}{q_0} \left( \frac{q_0}{q_f} \right)^{\frac{1}{b}}$$

$$= \frac{800000}{400} \left( \frac{400}{30} \right)^{\frac{1}{1}}$$

$$= 2666.67 \text{ yrs}$$

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2013/2014.

Recall Profit = Net operating income.

| Year | Production<br>bbls | Revenue<br>\$ | Operating<br>Cost \$ | UDP<br>\$ | Net Operating<br>Income BT<br>\$ | 46% tax<br>\$ | Profit AT.<br>\$ |
|------|--------------------|---------------|----------------------|-----------|----------------------------------|---------------|------------------|
| 1.   | 1,000              | 20,000        | 7500                 | 2285.714  | 10214                            | 4698          | 5516             |
| 2.   | 950                | 19000         | 8000                 | 2171.429  | 8229                             | 4061          | 4768             |
| 3.   | 900                | 18000         | 8500                 | 2057.143  | 7443                             | 3424          | 4019             |
| 4.   | 850                | 17000         | 9000                 | 1942.857  | 6057                             | 2786          | 3271             |
| 5.   | 800                | 16000         | 9500                 | 1828.571  | 4671                             | 2149          | 2522             |
| 6.   | 750                | 15000         | 10000                | 1714.286  | 3286                             | 1512          | 1774             |
|      |                    | \$105,000     |                      |           |                                  |               | \$21,276         |

Recall, Investment = \$12,000.

Recall,  $UDP = \frac{\text{Revenue in that year}}{\text{Total Sum of revenue}} \times \text{Investment}$ .Revenue = Production in bbls  $\times$  unit price \$20/bbl.Net operating Income BT =  $\$ (\text{Revenue} - \text{Operating Cost} - UDP)$ .

Net operating Income AT = Net operating income BT - tax.

tax =  $\frac{46}{100} \times \text{Net operating income BT}$ .

Hence, the total profit discounted from the investment = \$21,276.

2. From;  $A = F \left( \frac{i}{(1+i)^n - 1} + i \right)$

$$F = \$100,000$$

$$i = 0.2, n = 10 \text{ yrs}$$

$$\therefore A = 100,000 \left( \frac{0.2}{(1+0.2)^{10} - 1} + 0.2 \right)$$

$$\therefore A = \$28852.276$$

|      | PV<br>\$ | drilling cost<br>\$ | completion<br>cost \$ | Net Reserved<br>PV \$ $X_i$ | $P_i X_i$ | $P_i (X_i - \bar{X}_x)^2$ |
|------|----------|---------------------|-----------------------|-----------------------------|-----------|---------------------------|
| 0.65 | 0        | 150,000             | —                     | -150,000                    | -97500    | 5221500000                |
| 0.10 | 500,000  | 150,000             | 300,000               | 50,000                      | 5000      | 902500000                 |
| 0.10 | 1000,000 | 150,000             | 300,000               | 550,000                     | 55000     | 16402500000               |
| 0.10 | 1500,000 | 150,000             | 300,000               | 1250000                     | 105000    | 81902500000               |
| 0.05 | 2000,000 | 150,000             | 300,000               | 1550000                     | 77500     | 98701250000               |

$$\therefore \bar{X}_x = \sum P_i X_i = \$145,000.$$

$$\sigma^2 = \sum P_i (X_i - \bar{X}_x)^2 = \$1138523750000$$

$$\therefore \sigma = \$1067016.284.$$

$W = 0$  with PV discounting.

$Z = -1.645$  with a 5% risk.

$$M = \$10MM.$$

$$B_1 = \frac{X_A}{1+w} = \frac{145,000}{1+0} = \$145,000$$

$$B_2 = \frac{F}{2M} \left( \frac{ZD}{1+w} \right)^2 = \frac{1}{2(10,000,000)} \left( \frac{-1.645 \times 1067016}{1+0} \right)^2$$

$$\therefore B_2 = \$154043.627$$

Hence, from general risk eqn:

$$C \leq (B_1 + B_2) \left[ 1 - \sqrt{1 - \left( \frac{B_1}{B_1 + B_2} \right)^2} \right]$$

$$C \leq (145,000 + 154043.627) \left[ 1 - \sqrt{1 - \left( \frac{145,000}{145,000 + 154043.627} \right)^2} \right]$$

$$\therefore C \leq \$37505.685$$

Hence Management should not pay more than \$37505.685 for the lease at 5% Risk.

Decision A.

| $p_i$ | $PV \frac{x_i}{x_i}$ | $p_i x_i$ | $p_i (x_i - \bar{x}_A)^2$ |
|-------|----------------------|-----------|---------------------------|
| 0.3   | 120                  | 36        | 1732.8                    |
| 0.3   | 80                   | 24        | 388.8                     |
| 0.4   | -40                  | -16       | 2222.4                    |

$$\bar{x}_A = \sum p_i x_i = 44.$$

$$\therefore \sigma^2 = \sum p_i (x_i - \bar{x}_A)^2 = 4944.$$

$$\therefore \sigma = 70.314.$$

Decision B.

| $p_i$ | $PV \frac{x_i}{x_i}$ | $p_i x_i$ | $p_i (x_i - \bar{x}_B)^2$ |
|-------|----------------------|-----------|---------------------------|
| 0.2   | 150                  | 30        | 2691.2                    |
| 0.4   | 60                   | 24        | 270.4                     |
| 0.4   | -50                  | -20       | 2822.4                    |

$$\bar{x}_B = \sum p_i x_i = 34.$$

$$\therefore \sigma^2 = \sum p_i (x_i - \bar{x}_B)^2 = 5784$$

$$\therefore \sigma = 76.053$$

### Decision

Since the branch with lower expected value (Mean  $\bar{X}_x$ ) and greater Variance ( $\sigma$ ) is trimmed off.

Hence, decision  $\beta$  with  $\bar{X}_x = 34$  and  $\sigma = 76.053$  is trimmed off.

| $P_i$ | Net Producing Interval (ft) | Net Production (bbl) | Net Operating Profit \$ | NPV operating profit \$ |
|-------|-----------------------------|----------------------|-------------------------|-------------------------|
| 0.4   | 0                           | 0                    | 0                       | 0                       |
| 0.2   | 15                          | 90,000               | 180,000                 | 126,000                 |
| 0.2   | 20                          | 120,000              | 240,000                 | 168,000                 |
| 0.2   | 25                          | 150,000              | 300,000                 | 210,000                 |

Production = 150 bbl/acre-ft

Spacing = 40 acre

PV factor = 0.7

Operating profit = \$2/bbl

$n = 20$

90% Confidence, 10% risk

$Z = -1.282$

$W = 50\% = 0.5$

| $p_i$ | $\$X_i$ | $p_i X_i$ | $p_i (X_i - \bar{X}_n)^2$ |
|-------|---------|-----------|---------------------------|
| 0.4   | 0       | 0         | $4.064 \times 10^6$       |
| 0.2   | 126,000 | 25,200    | $127 \times 10^6$         |
| 0.2   | 168,000 | 33,600    | $903 \times 10^6$         |
| 0.2   | 210,000 | 42,000    | $2384 \times 10^6$        |

$$\bar{X}_n = \$100,800$$

$$T^2 = \sum p_i (X_i - \bar{X}_n)^2 = 7478 \times 10^6$$

$$\therefore T = \$86475$$

$$\therefore C \leq \frac{1}{1+w} \left( \bar{X}_n + \frac{T Z}{\sqrt{n}} \right)$$

$$C \leq \frac{1}{1+0.5} \left( 100,800 + \frac{(-1.282 \times 86475)}{\sqrt{20}} \right)$$

$$C \leq 0.667 \left( 100800 + \left( \frac{-11086.95}{4.4721} \right) \right)$$

$$\therefore C \leq \$50699.0035$$

Hence, the Company should not invest more than \$50699.0035 in the work over.

Given:  $N_{pD} = 800,000 \text{ bbl}$   
 $q_0 = 400 \text{ bbl/d}$   
 $q_f = 30 \text{ bbl/d}$

for exponential decline  $b=0$ .

$$t_f = \frac{N_{pD}}{q_0 - q_f} \ln \left( \frac{q_0}{q_f} \right)$$

$$= \frac{800,000}{400 - 30} \ln \left( \frac{400}{30} \right)$$

$$\therefore t_f = 5600.578 \text{ days}$$

for hyperbolic decline with  $b=0.5$ .

$$t_f = \frac{N_{pD}}{q_0} \sqrt{\frac{q_0}{q_f}}$$

$$= \frac{800,000}{400} \sqrt{\frac{400}{30}}$$

$$\therefore t_f = 7.30 \text{ days}$$

iii. For Harmonic decline with  $b = 1$ .

$$t_f = \frac{N_{pD}}{q_0} \frac{(q_0/q_f - 1)}{\ln(q_0/q_f)}$$

$$= \frac{800,000}{400} \frac{(400/30 - 1)}{\ln(400/30)}$$

$$\therefore t_f = 9522.827 \text{ days.}$$

## Paper 2 of 2 — PEE 513 (Petroleum Properties Evaluation & Management)

Level 500 • First Semester • Handout

PETROLEUM PROPERTIES

EVALUATION & MANAGEMENT

PEE 513

By Dr. Taiwo O. A

Probability Distribution

- Probability is a measure of uncertainty about the occurrence of a particular event.
- It indicates our degree of belief that a certain event will occur.

In everyday language, it is expressed thus:

- There is a high chance that it will rain tonight
- It is possible that the super eagles will lose to their opponents.

As long as we have uncertainty, we can never be sure what the outcome of any venture or business decision will be.

Probabilities help us to factor into our decisions these uncertainties and used intelligently, improves the quality of our decision making.

Basic probability theory

There are generally two broad classes of probabilities –

Objective and Subjective Probability. The Modern approach combines both Objective and Subjective concepts of probability

Basic Probability Theory

- Objective probability – Logical deduced or empirically observed through experimentation
- It is not based on the opinions of experts or any individuals.
- Example - toss of a fair coin, throw of a die (dice)
- Subjective probability - Based on the opinions of experts using experience and intuition.
- Here the degree of belief definition of probability is dominant
- Example: a Geologist's assessment of the probability of dry hole in a drilling venture=0.5

Basic Probability Theory - Concepts

- Probabilities are always greater than or equal to zero and always less than or equal to one
- If events are mutually exclusive and collectively exhaustive, their probabilities must add up to one
- Probabilities can be expressed as fractions or percentages.

Consider the toss of 2 fair coins

The following are the possible outcomes or Sample Space = {HH, HT, TH, TT}

Probability of each outcome (events) is 0.25 or 25%

$P(HH) = 1/4$ ,  $P(HT) = 1/4$ ,  $P(TH) = 1/4$ ,  $P(TT) = 1/4$ : Total = 1.00

Basic Probability Theory - Concepts

- Independence - If two events A and B are statistically independent, the occurrence of one event has no influence whatsoever on the occurrence of the other.
- Thus:  $P(A) = P(A|B)$  and  $P(B) = P(B|A)$
- Dependence - Above relationships do not hold.  $P(A) \neq P(A|B)$ , The occurrence of event A, has some influence on the likelihood of the occurrence of B.
- Example : Prob (Oil|Favorable Seismic) or Prob (Commercial Reserves|Discovery)

Basic Probability Theory - Concepts

Simple Probability -  $P(A)$  - Probability of occurrence of an event

independent of any other event - Also called independent probability.

-  $P(A)$  is the probability of the event A occurring regardless of any other event

Conditional Probability -  $P(A|B)$  - Measure of an event, A occurring given the occurrence of another event, B. Example :  $P(\text{Commercial Reserves} | \text{Discovery})$  is the probability of commercial reserves given that we have made a discovery.

Joint Probability -  $P(AB)$  - is the probability of events A and B occurring simultaneously or in succession.

Example : if the probability of a producer is 0.6 and the probability that the producer will have reserves of 50 million barrels is 0.25, then the probability that reserves will be 50 million barrels is 0.15 (0.6 times 0.25)

Marginal probability - Sum of Joint Probabilities

Basic Probability Theory - Rules

Complementation

$P(A) + P(B) = 1$  (since the sum of probabilities of collectively exhaustive events = 1)

Where  $P(A)$  = Probability of event A  $P(B)$  = Probability of not event A

Multiplication  $P(AB) = P(A \text{ and } B) = P(A) \times P(B)$  For two events that are independent, their joint probability is a simple multiplication of their two probabilities). Example : Throw of 2 fair coins,  $P(HH) = P(H) \times P(H) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$  Addition  $P(A \text{ or } B) = P(A) + P(B)$  For two mutually exclusive events, the sum represents the various different ways that an outcome can occur.

Example :  $P(HT \text{ or } TH) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$  Multiplication  $P(AB) =$

$P(A \text{ and } B) = P(A) \times P(B)$  For two events that are independent, their joint probability is a simple multiplication of their two probabilities). Example :

Throw of 2 fair coins,  $P(HH) = P(H) \times P(H) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Addition  $P(A \text{ or } B) = P(A) + P(B)$  For two mutually exclusive events, the sum represents the various different ways that an outcome can occur. Example :

$P(HT \text{ or } TH) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Theoretical distributions

Discrete Distributions -examples

Binomial, Negative Binomial, Multinomial, Hyper geometric

Continuous Distributions forms -examples

Exponential, Poisson, Beta, Gamma, Uniform, Triangular,

Normal, Lognormal.

Theoretical distributions - Discrete

- Binomial Distribution
- Discrete Distribution – suitable for situations in which we have two outcomes viz Head or Tail, Dry Hole vs Producer, Success or Failure
- Represents sampling with replacement and independent trials- e.g. early exploratory stage

Probability of having x successes in n trials is given thus:

Where n = no of trials p = probability of success q = probability of failure = 1- p x = exact no of successes

Theoretical distributions - Discrete

- Example 1: What is the probability of at least a (1) head in three throws of a fair coin? This is the complement of no Heads at all in all three throws or  $1-p(x=0, n=3)$ . In this example, the no of trials is 3, the number of successes is 0 and the probability of success is  $\frac{1}{2}$ .

Theoretical distributions - Discrete

- Example 2 – Gamblers’ Ruin Determination using the Binomial Distribution

• Gamblers’ Ruin is defined as the chance of “run of bad” luck or bad outcomes in a series of gambles leading to the entire loss of investment capital or bankruptcy.

• Bankruptcy denotes a situation in which the enterprise folds up, goes into liquidation and is unable to continue to conduct business which would have led to the recovery of prior losses.

• In an Oil exploration drilling program, Gamblers’ Ruin will occur when all the wells drilled turn out to be dry holes leading to loss of the entire exploration budget.

Theoretical distributions - Discrete

Example: Let’s say Oil Company wishes to execute a 5-well exploration program costing \$5 million. If

each well costs \$1 million, then the Company can drill 5 wells with the budget of \$5 million. What is the probability of Gambler's Ruin or the chance that all 5 wells will turn out to be dry? The probability of success (or finding a discovery) in this Oil province is 20% (say, historically and we believe this has not changed) .

Theoretical distributions - Discrete

Example, continued: Suppose the Company invites a partner to co-fund the project with an additional \$5 million dollars so that each participating company now has 50% of the venture. The number of wells that can be drilled is now 10. What is the chance of Gambler's ruin in the new exploration program in which the risks of investment is shared equally between the two Companies?

The Gambler's ruin probability has dropped to 10.74% for the 10-well venture as against 32.8% for the 5 well venture in which the 1st Company was solely "carrying" all the risk of the exploration program. This form of risk sharing is common in the Upstream Oil & Gas business - by increasing the number of wells through risk sharing,

Theoretical distributions - Negative binomial Distribution

The Negative Binomial Distribution is used to estimate the number of failures that will occur before success is attained.

where  $x$  = number of failures met before achieving the  $r$ th success  $r$  = required number of successes  $p$  = probability of success in each trial  $q = 1 - p$  = probability of failure on each trial

Example: An Oil Independent is about to embark on an exploration program in a new basin. The Company

plans to drill ten wells at \$1 million dollars each. The total budget for the exploration program is \$10 million.

The historical probability of success in this Exploration basin is 0.2 and is deemed not to have changed as a

result of prior programs in the area. Management wants to know the following before approving the exploration program:

1. What is the chance that all the Exploration wells will turn out to be dry holes?
2. What is the chance that there will be two dry holes before a discovery is made?
3. What is the probability of drilling 5 dry holes before a discovery is made?

The answer to the first question can be estimated in either of two ways by using the Binomial distribution or the Negative Binomial.

Theoretical distributions - Binomial/Negative binomial

Binomial distribution Solution:

Using the Binomial distribution, no of successes,  $x = 0$ , no of trials = 10 and probability of success is 0.2.

Theoretical distributions - Multinomial Distribution

This is a discrete distribution which unlike the Binomial distribution has many outcomes. For instance, discovered reserves in a new exploration basin may range

from 0, 10 Million, 15 Million barrels. The probabilities of all these outcomes all sum up to 1.

Where,  $P(S)$  is probability of particular sample  $P_1, P_2, \dots, P_m$  are probabilities of drawing types 1, 2, ...,  $m$  respectively from the population

$N = k_1 + k_2 + \dots + k_m$  is the total number of outcomes of types 1, 2, ...,  $m$  respectively in sample  $m$  is the total number of different types of outcomes within the total sample.

Theoretical distributions – Multinomial Distribution

Example: Suppose the probability of dry hole, 12 MMSTB and 18 MMSTB Reserves in the figure above are

0.50, 0.35 and 0.15 respectively. A Company plans to drill three wells, what will be the probabilities of

discovering various total reserves with the three additional wells?

The various possible combinations of the three wells are as follows: all three may turn out to be dry and all

three may turn out to be discoveries of 18 MMBls each (0 and 18 MMBls being the two extremes of possible

Reserves and various combinations in between as listed in the table below):

For  $k_1 = 2, k_2 = 1, k_3 = 0$  combination, the corresponding probability is:  $m = 3, N = 3$

Theoretical distributions - Continuous

Uniform Distribution: The uniform distribution is the “coarsest” form of distribution implying very scant information of the variability of the random variable,  $x$  in question. The distribution is stipulated only by the range or the minimum and maximum values of  $x$  and the assumption that all other values of  $x$  are equally probable.

Theoretical distributions – Continuous- Uniform

Example: The operating cost of a centrifugal compressor station ranges between \$5,000 and \$10,000 per month, with the values between these minimum and maximum values all equally probable.

- What is the mean cost of operating this pumping station per month?
- What is the probability density function for this distribution?
- What is the chance that the operating will be less than \$8,000/month? Greater than \$6000/month?

Theoretical distributions – Continuous- Triangular

The triangular distribution function is very popular in its application due to the fact that people can easily relate to the maximum, minimum and most likely values of a random variable.

Theoretical distributions

Expected Value

Expected value of a real valued variable gives the centre (or mean) of the distribution of the variable

- Referred to in several different ways viz: Expected Value, Expectation Value, Mathematical expectation, Mean of a random variable.

Example

Expected value of a roll of die – the outcomes are 1, 2, 3, 4, 5 & 6 each with a probability of 1/6. Expected Value,  $EV = 1/6(1)+1/6(2)+1/6(3)+1/6(4)+1/6(5)+1/6(6)=21/6 = 3.5$

Expected Value

Oil Field Example •

A well treatment will cost \$50,000 whether or not it is successful. The following are the various possible increases in the present value of production which may be realized as a result of the proposed treatment. Calculate the net present value of each outcome, and the mean and standard deviation of the possible net present value outcomes.

Expected Value

We will only calculate the mean or expected value (EV) at this stage. The calculation of the standard deviation, we will leave till when we define that term.

Oil Field Example-Solution • We are told the well treatment will cost \$50,000 whether or not it is successful. If we define net present value as:  $NPV = \text{Present Value (PV)} - \text{Investment, I}$  Then we can calculate the NPV for each of the outcomes.  
 $NPV (PV=0) = 0 - \$50,000 = -\$50,000$

$EV = (0.25)(-50,000) + (0.10)(-10,10,000) + (0.30)(+10,000) + (0.20)(+30,000) + (0.15)(+50,000)$  Expected Value,  $EV = \$3,000$

Expected Value

Oil Field Example-Solution shown graphically – as part of decision tree.

Recall Expected Value, EV is defined as the mean or average of a series of random variables (real).

It is used to predict the outcome of an experiment or trial or an uncertain situation like a drilling situation. It's usefulness is significant if the outcome of the uncertain event does not deviate too much from the Expected Value

Discounted Cash Flow

Analysis

Discounted Cash flow Analysis

The Concept of Interest

- The interest is the amount charged on top of the principal by a lender to a borrower for the use of assets.
- It is the price or rent paid for the use of money, expressed as a percentage of the loaned or borrowed fund.

The interest rate is the cost of debt for the borrower and the rate of return for the

lender

- Interest rate could be risk-free, market risk premium,
- There are two major sources of financing for business firms: equity financing (through stocks) and financing through debts (loans from banks, credit unions, investment bankers, etc)
- In this case, the interest rate is referred to as Weighted average cost of capital (WACC), computed from the weighted average of cost of loan and equity.

Discounted Cash flow Analysis

Where • E=Market value of the firm's equity

- D=Market value of the firm's debt
- V= Total Market Value =E+D
- Re=Cost of equity
- Rd=Cost of debt
- Tc=Corporate tax rate

Example: Calculate the WACC of an investment project if 20% of the project is financed by equity at a cost of 15% per annum and 80% is financed by loan borrowed at a rate of 18% per annum given at corporate rate of 30%

Solution

$$\text{WACC} = (20/100 \times 0.15) + (80/100 \times 0.18 \times (1-0.3)) = 0.03 + 0.1008 = 0.1308$$

Discounted Cash flow Analysis

The Concept of Interest

Interest rate application to loan may be simple or compound. However, the compound interest rate is usually applied.

Interest= Amount (A)- Principal (P)

Amount is expressed as

Interest is expressed as

Where i=nominal interest rate

M= compounding interest/yr

n=loan period or investment period.

Discounted Cash flow Analysis

Nominal and Effective Interest Rate

Compound interest is normally reported as nominal interest rate per year, compounded periodically (i.e., daily, monthly, quarterly, or semiannually) or continuously (infinite number of times per year, approximately equal to daily compounding).

- Effective interest rate is the rate that when applied to a sum of money once per year will give the same amount of interest as will be given by a nominal interest rate compounded m periods per year.

Discounted Cash flow Analysis

Example 7: Suppose a sum of \$2,000 is borrowed for one year at an interest rate of

8%. Calculate the total amount to be paid back to the bank after five years if the interest is compounded quarterly and effective interest rate.

Solution

Time Value of Money • The concept of interest is the key to explaining time value of money. A dollar is worth more in the future. In accounting, the principal (PV) is referred to as the Present value of loan or investment while amount is referred to as the future value (FV)

Discounted Cash flow Analysis

Time Value of Money

\* Equal Series of Payment This is called annuity expressed as:

Where PV=Present Value FV= Future Value n=Number of years i=Discount rate.

Time Value of Money is due to: • Interest Rate • Inflation • Risk • Opportunity Cost •

Security

Discounted Cash flow Analysis

Capital Expenditure (CAPEX)

• CAPEX is expenditure for physical assets such as land, property, plants, equipment • It may tangible or intangible CAPEX

• Tangible Assets are those of physical nature such as lease tanks, wellheads, buildings, etc. They have a useful of more than one year and recognised salvage value. Tangible assets are usually depreciated.

• Intangible Assets lack physical substance. These include oil and gas leases, a portion of drilling costs expensed on chemicals, consumables; patents, rights. Intangible assets are usually amortized.

• Depletion applies to progressive removal or depletion, of natural resources.

• Collectively these write-offs are frequently called DD&A

Discounted Cash flow Analysis

Operating Expenses (OPEX)

OPEX are expenditures that occur periodically

• They are necessary for the day-to-day operations of the field.

• They are usually expressed in terms of expenditure per year, per barrel, or per MScf

They include:

• Feed Stock • Utilities • Maintenance • Administrative Overhead • Production cost

General Rule of Thumb for OPEX

Discounted Cash flow Analysis

Loan Amortisation Schedule

• Loan amortisation means paying off a debt, normally a gradual retirement of debt with a series of equal periodic payments. Depending on the type of loan repayment arrangements, the series of payments will normally include interest accrued on the outstanding loan balance and some portion of the principal.

Example: • An oil producer borrows \$100,000 at an interest rate of 8% per year for a

period of three years. The loan has to be paid back in equal quarterly payments over the three-year period, with the first payment due exactly three months from the date money is borrowed. Calculate the quarterly payment and show the loan amortization schedule.

Discounted Cash flow Analysis

Depreciation

Depreciation: Loss in the value of an asset over its useful life Events that can cause property to depreciate include wear and tear, age, deterioration, and obsolescence.

Book Value: Value of an asset at anytime during its useful life

Salvage Value: Value of an asset at the end of its useful life.

Discounted Cash flow Analysis

Discounted Cash flow Analysis

Example: Sum-of-Years'-Digit (SYD) Compute the annual depreciation and book value of an asset over

its useful life of 5 years if the initial cost of the asset is \$35,000 and salvage value at the end of its useful life is 0, using SYD method.

Discounted Cash flow Analysis

Project Economic Valuation

Economic Yardstick Decision Criteria

Net present value (NPV)

Internal rate of return (IRR)

Payback period

Profit-Investment-Ratio

Present Value Ratio

Profitability index

Cost benefit ratio

Unit technical cost or long-run marginal cost

Discounted Cash flow Analysis

Net Present Value

- The net present value (NPV), also referred to as the present value of cash surplus or present worth,

is obtained by subtracting the present value of periodic cash outflows from the present value of periodic cash inflows.

- The present value is calculated using the weighted average cost of capital (WACC) of the investor, also referred to as the discount rate or minimum acceptable rate of return. Equation for NPV is

Where:  $NCF_t$  = Net Cash flow at time  $t$

$i$  = the discount rate, i.e., the required minimum annual rate of return on new investment

$t$  = Period, year  $n$  = the project's economic life in year

Discounted Cash flow Analysis

Internal Rate of Return (IRR)

- IRR is the discount rate at which the net present value is exactly equal to zero, or the present value

of cash inflows is equal to the present value of cash outflows.

- IRR is the interest rate received for an investment consisting of payments (negative values) and income (positive values) that occur at regular periods.
- The equation for calculating IRR is:

Discounted Cash flow Analysis

Payback Period

- The payback period, also referred to as the breakeven point, is defined as the expected number of

years required for recovering the original investment.

- At this point, the cash receipts exactly equal the cash disbursements. This yardstick is used along with at least one other measure of profitability, since it does not provide a meaningful decision criterion by itself.

Formula for computing Payback period:

Discounted Cash flow Analysis

Profit-Investment-Ratio-PI is the ratio of the undiscounted NCF and undiscounted Investment. As a rule of thumb, when the PI is at least 1.00 the project is likely to be economic.. The unit is dimensionless or \$/\$ each dollar return for each dollar invested. It is a measure of investment efficiency. Especially popular with small to medium sized firms.

Present Value Ratio - PVR is the ratio of the discounted NCF and discounted Investment at a discount

rate. A variant of this is the

Discounted Profit-Investment-Ratio (DP/I) which is the ratio of the discounted NCF and discounted "after tax" Investment.

Profitability Index (PI) PI is a dimensionless ratio obtained by dividing the present value of future operating cash flows by the present value of the investment. Mathematically, the PI is given by the following equation:

Discounted Cash flow Analysis

Example: A capital investment project involves an initial investment of \$1,000, financed 75% by equity and 25% by debt. The corporate tax rate is 35%, interest on the debt is 6.15%, and the minimum acceptable rate of return on equity is 10% (assuming the equity can be invested elsewhere

at a rate of return of 10%). The project will generate after-tax operating cash flows as shown in table

1. Calculate the NPV, IRR, and PI for the investment

Discounted Cash flow Analysis

Decision Trees

- A decision tree reflects a decision problem having more than one course action. It is a diagrammatic representation of a decision situation.

- The decision trees can help a decision maker to develop a clear view of the structure of a problem and make it easier to determine the possible scenarios that can result if a particular course of action is chosen.
- Decision trees depict pictorial views of the available courses of action, events, their outcomes, and probabilities associated with the outcomes.

#### Decision Trees

- Hence, elements of decision trees include:
- Decision node: Point where decision has to be taken. The decision maker has control at this point
- Chance node: Point where there are different possible outcomes. The decision maker has no control. Outcomes are dependent on chance or nature
- Chance fork: The branch from a chance node is called chance fork and it represents the possible outcome of a chance event.
- Probability or Chance: The likelihood of possible outcomes happening is represented by probabilities in decision trees
- End Terminal or payoff node: It is deterministic financial outcome of a decision. It is based on predicted profit or loss of an outcome and the probability of occurrence

#### Decision Trees – two-activity Decision Tree

Expected value  $E\{C1\}$  of the chance node C1 is the sum of the product of the conditional probability  $P_i$  and associated computed NPV<sub>i</sub> at each end node.

At each chance node, probabilities must sum up to 1

- Decision trees are solved by rolling them back from right to left
- The end nodes show the pay-offs; if it is a dry hole, loss is \$65,000 (Drilling cost), while NPVs are \$120,000 and \$180,000 for reserves of 60 Mstb and 90 Mstb, respectively.

#### Decision Trees

##### Example

A company has to decide whether to drill a certain prospect. If it drills the prospect, geologists and engineers expect

- probability of zero reserves (dry hole) to be 60%,
- probability of 60,000 barrels to be 30%, and
- probability of 90,000 barrels to be 10%. The net present value for

each option is calculated as shown in the decision tree above. Based on this information, determine the optimum course of action the company should take in order to maximize its EMV.

Decision Trees - solution

Decision Trees

Example 2

A company is planning to drill a well. Geologists and engineers estimate that there is a 65% chance that the well will be a producer and a 35% chance that it will be a dry hole.

If the well is successful, it is estimated that there is a 60% chance that it will have reserves of 30,000 barrels, a 30% chance of 60,000 barrels, and a 10% chance of 90,000 barrels.

The dry hole cost is \$65,000 and the NPV corresponding each reserve value is \$60,000, \$120,000 or \$180,000, respectively.

Using decision tree, calculate the expected monetary value (EMV) for this proposal.

Decision Trees - solution

Decision Trees - solution

- By utilizing joint probabilities

Decision Trees – practice exercise

- There are three field development options (1) continue with primary depletion, (2) implement waterflood, or (3) implement gas injection.

The net present value at the corporate discount rate and corresponding probabilities of realizing these NPVs are given in the table that follows. Using this data, perform expected value analysis to arrive at the most economically viable option. Base decision on 1.

EMV 2. EPI

Decision Trees – Bayesian rule

- Bayesian Probability

Bayesian rule states that:

Decision Trees

- Steps to applying Bayesian rule using Decision Tree:

Construct the tree with all possible events on the branches indicating all prior probabilities for the events on the branches

- Attach a new branch to each branch representing information to be obtained. E.g. 90% chance of acquisition of seismic will indicate structure when structure actually exist, and there is 10% chance it will indicate structure when structure actually does not exist. This additional information leads to additional branches of probability tree.
- Obtain the joint probabilities by multiplying each prior probability by

the conditional probability that follows it on the tree.

- Sum the joint probabilities.
- Divide each joint probability by the sum of the joint probabilities to obtain the required posterior probability

Decision Trees – Bayesian example

- A geologist has assigned a 70–30 chance for a prospect to be gas bearing (70% chance that gas exists and 30% chance that gas does not exist). A well is drilled and tested. The test confirms the existence of gas. However, based on experience, it is believed the test is reliable only 90% of the time. This means 10% of the time the test shows existence of gas while the gas does not actually exist.
- Based on this information, revise the prior probabilities

Decision Trees – Bayesian example solution

- Prior Probability
- $P(\text{Gas})=0.7$ ,  $P(\text{No gas})=0.3$
- Test Probability
- $P(\text{test}>\text{gas}|\text{gas})=0.9$ ,  $P(\text{test}>\text{gas}|\text{no gas})=0.1$
- $P(\text{test}>\text{no gas}|\text{no gas})=0.9$ ,  $P(\text{test}>\text{gas}|\text{no gas})=0.1$

Decision Trees – Bayesian example solution

Decision Trees – Bayesian example solution